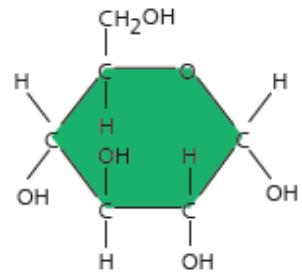
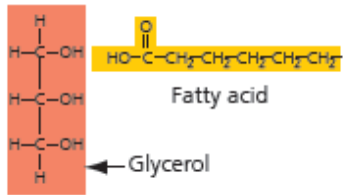
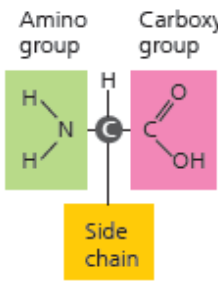
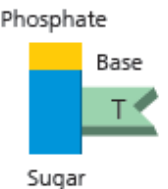


The Molecules of Life

Unit 1; Chapter 3

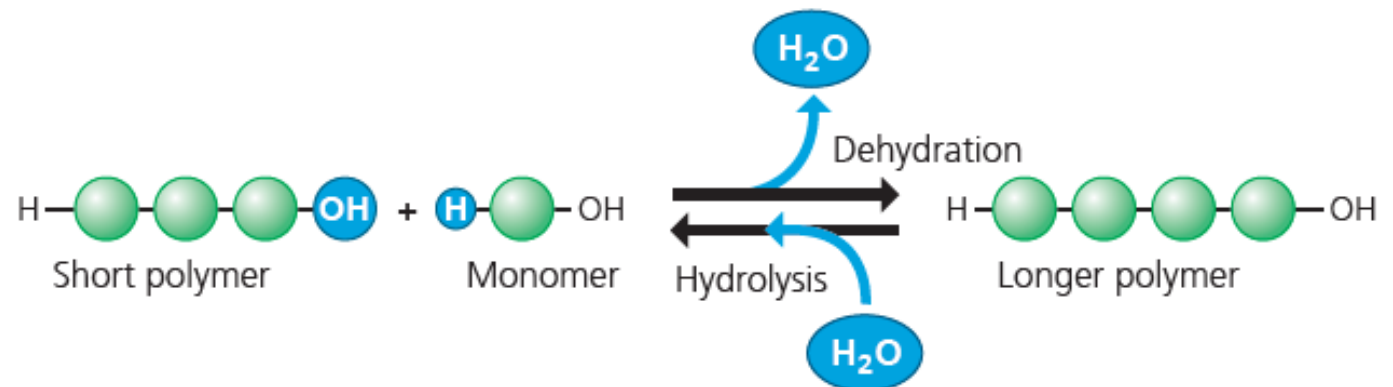


Large Biological Molecules	Functions	Components	Examples
Carbohydrates	Dietary energy; storage; plant structure	 Monosaccharide	Monosaccharides: glucose, fructose; disaccharides: lactose, sucrose; polysaccharides: starch, cellulose
Lipids	Long-term energy storage (fats); hormones (steroids)	 Fatty acid Glycerol Components of a triglyceride	Fats (triglycerides); steroids (testosterone, estrogen)
Proteins	Enzymes, structure, storage, contraction, transport, etc.	 Amino group Carboxyl group Side chain Amino acid	Lactase (an enzyme); hemoglobin (a transport protein)
Nucleic acids	Information storage	 Phosphate Base Sugar Nucleotide	DNA, RNA

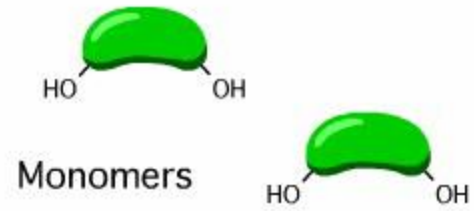
Giant Molecules from Smaller Building Blocks

- LO: Compare a **dehydration** (脱水) reaction to a **hydrolysis** (水解) reaction.
- **Macromolecules** are **polymers** (聚合物).
 - Carbohydrates, Proteins, Nucleic acids
- Polymers are made by stringing together many smaller molecules called **monomers** (单体).

Giant Molecules from Smaller Building Blocks



Animation: Polymers



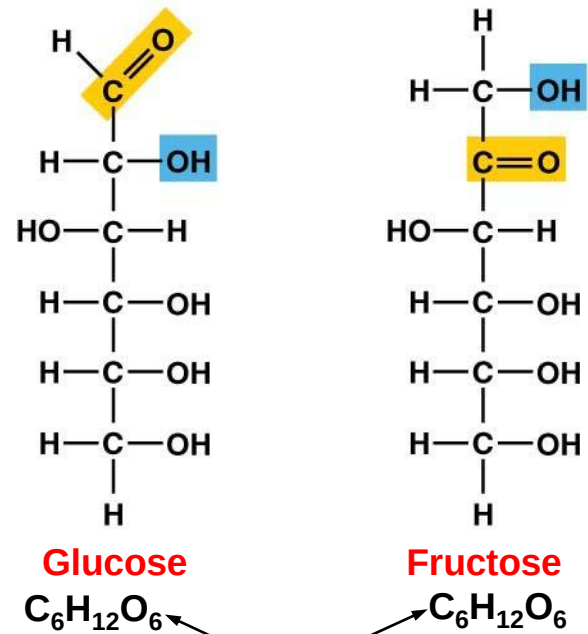
Carbohydrates (碳水化合物)

- LO: Compare the structures and roles of monosaccharides, disaccharides, and polysaccharides in living organisms. Give examples of each.
- Include sugars and polymers of sugar:
 - small sugar molecules in soft drinks
 - long starch molecules in spaghetti and bread
- **Mono**saccharides, **Di**saccharides, **Poly**saccharides
- In animals
 - a primary source of dietary energy
 - raw material for manufacturing other kinds of organic compounds
- In plants, as a building material for much of the plant body

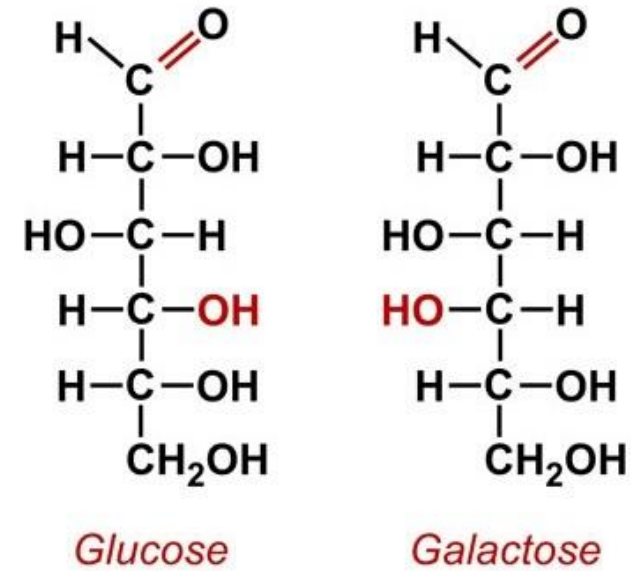
Monosaccharides (单糖)

- Monomers of carbohydrates
- Cannot be broken down into smaller sugars
- General formula $(CH_2O)_n$, where n is usually 3, 4, 5, 6, 7, or 8
- **Glucose** (葡萄糖) in sports drinks
- **Fructose** (果糖) found in fruit.
- Glucose and fructose are **isomers** (同分异构体), molecules that have the same molecular formula but different structures.

Fructose tastes about 1.7 times sweeter than glucose.



Isomers
(same formula, different arrangements)



Galactose is only about half as sweet-tasting as glucose.

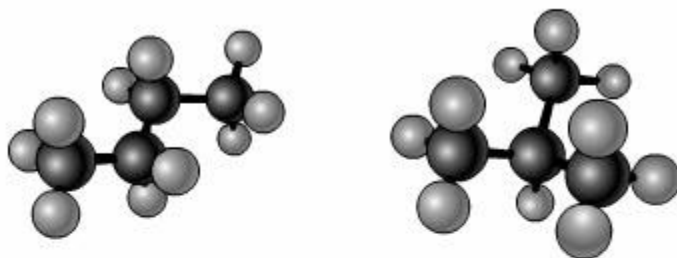


Celery

Animation: Isomers

结构异构

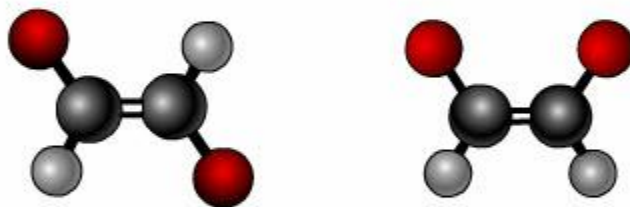
Structural isomers



几何异构

Geometric isomers

(Cis-trans isomers)



对映异构

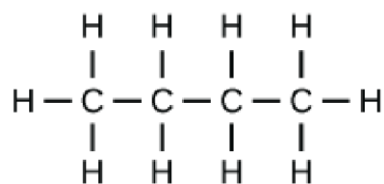
Enantiomers

Stereoisomers

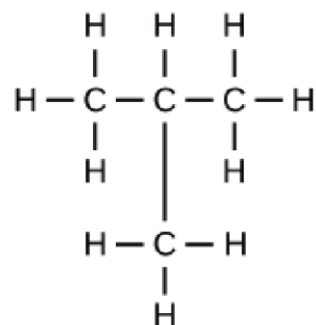


Structural isomers

Butane

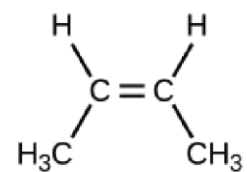


Isobutane



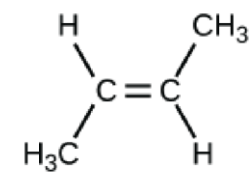
Geometric isomers

cis-2-butene



methyl groups on
same side of double bond

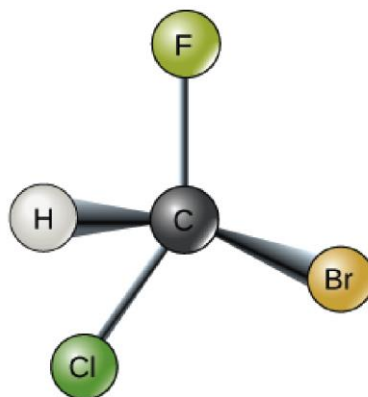
trans-2-butene



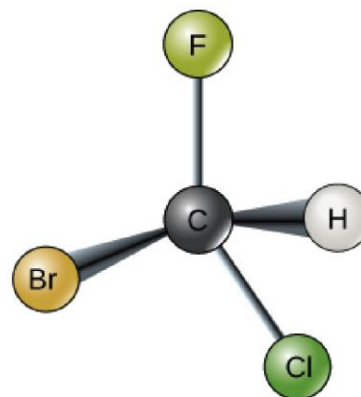
methyl groups on opposite
sides of double bond

Enantiomers

L-isomer

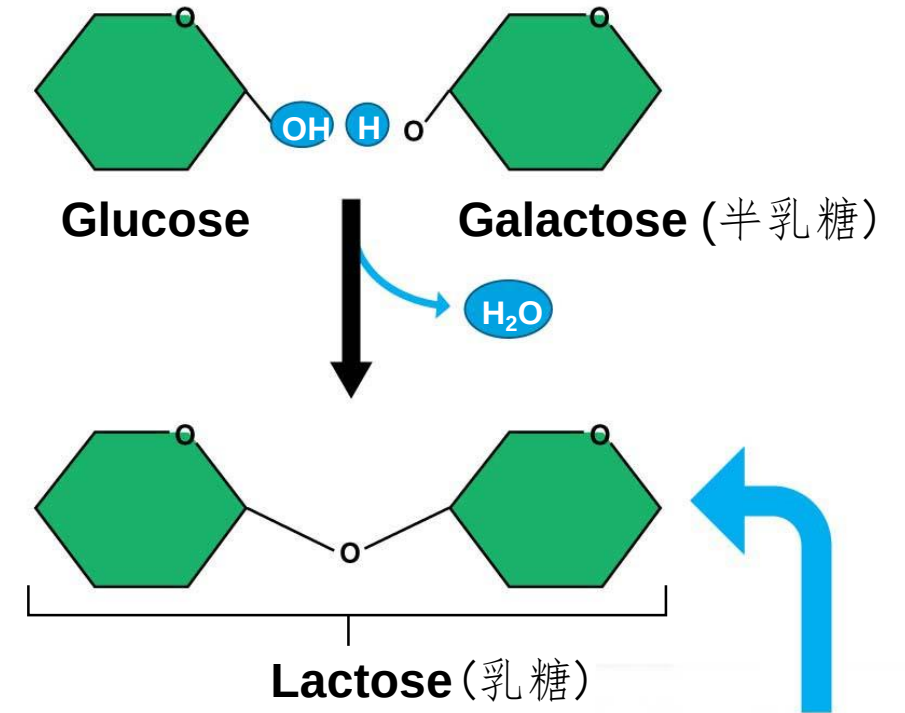


D-isomer



Disaccharides (双糖)

- Double sugar constructed from two monosaccharides by a dehydration reaction
- Lactose (乳糖) in milk, breast milk
- Maltose (麦芽糖) in beer
- Sucrose (蔗糖) in table sugar
 - A major cause of tooth decay
 - Overconsumption increases the risk of developing type 2 diabetes (糖尿病) and heart disease

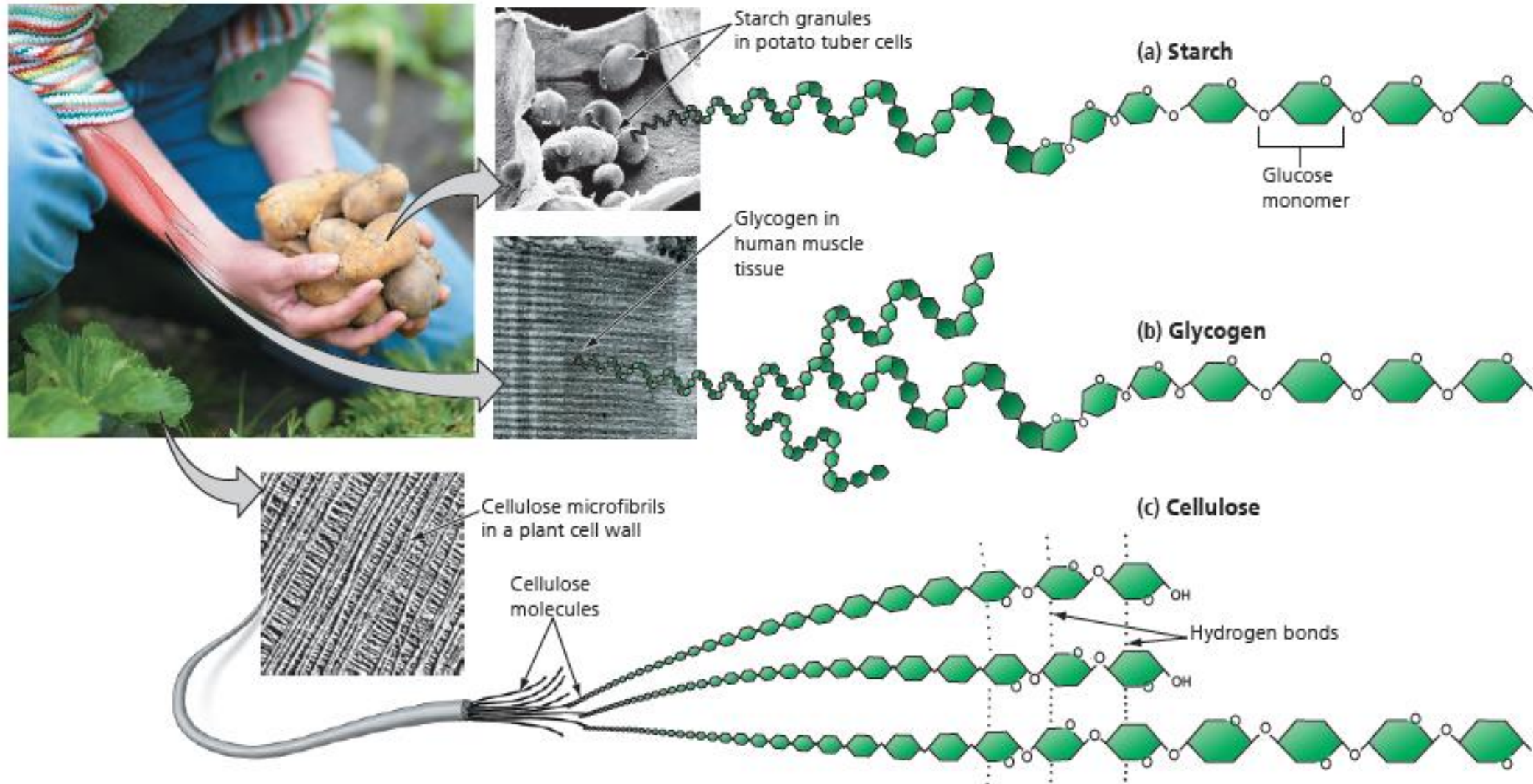




Lactose Intolerance (乳糖不耐受)

Polysaccharides (多糖) are Complex carbohydrates made of long chains of sugars—polymers of monosaccharides.

▼ **Figure 3.9** Three common polysaccharides.

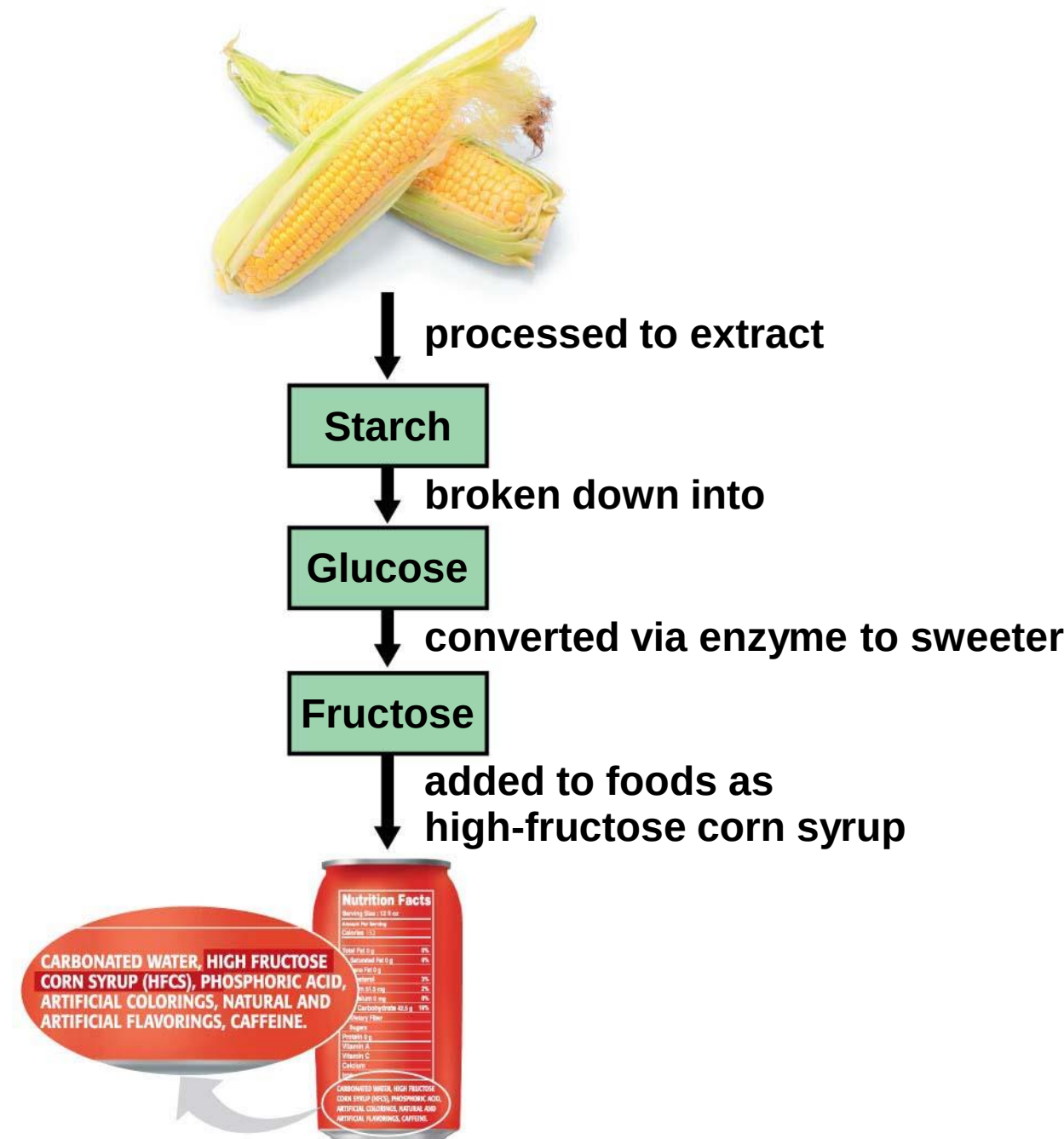


Polysaccharides (多糖)

- Starch (淀粉)
 - Used by plant cells to store energy
 - Potatoes and grains in our diet
- Glycogen (糖原)
 - Used by animal cells to store energy in muscle and **liver**
 - broken down to release glucose when you need energy
- Cellulose (纤维素)
 - Forms cable-like fibrils (原纤维) in the walls that enclose plant cells
 - The **most abundant organic compound** on Earth
 - Cannot be broken by any enzyme produced by animals

High-fructose corn syrup (HFCS) (高果葡糖浆)

- Common sweetener
- HFCS is often one of the first ingredients listed in soft drinks.
- 55% fructose
- To replace sucrose
- Why not sucrose?



Lipids (脂类)

LO:

- Describe the unique properties of lipids.
- Compare the structure and properties of saturated and unsaturated fatty acids.
- Distinguish between steroids and anabolic steroids and explain how the use of anabolic steroids can be dangerous to a person's health.

Hydrophilic and Hydrophobic

- Hydrophilic (“water-loving”)
 - Philip likes to drink water
 - Dissolve readily in water
 - Carbohydrates, amino acids, nucleotides
- Hydrophobic (“water-hating”)
 - Lipids
 - Unable to mix with water
 - “Hydrophobic force”?



Lipids

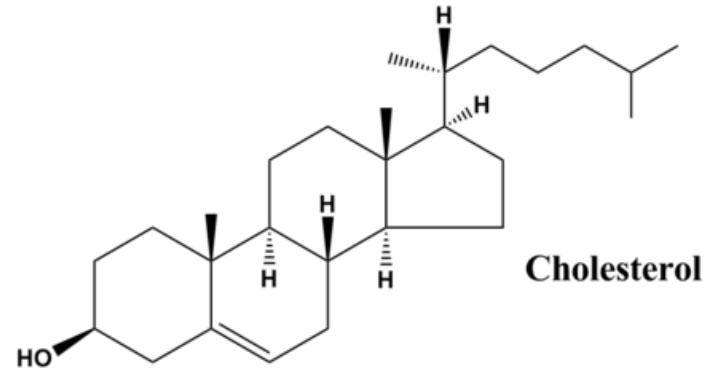
- Lipids are a **diverse** group of molecules made from different molecular building blocks.
- Large (not giant) biological molecules.
- Not macromolecules (polymers built from repeating monomers)
 - Carbohydrates, by monosaccharides
 - Proteins, by amino acids
 - Nucleic acids, by nucleotides

Diverse Lipids

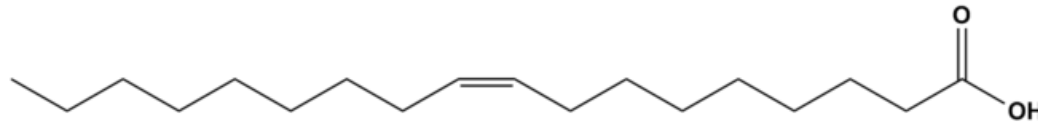
- Fats

- Steroids

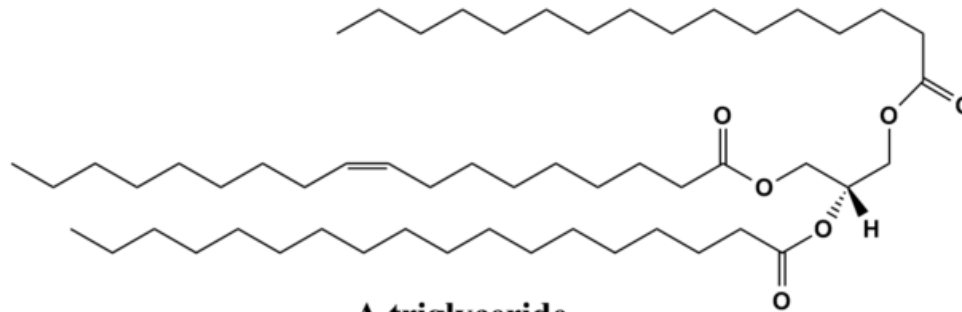
- Phospholipids



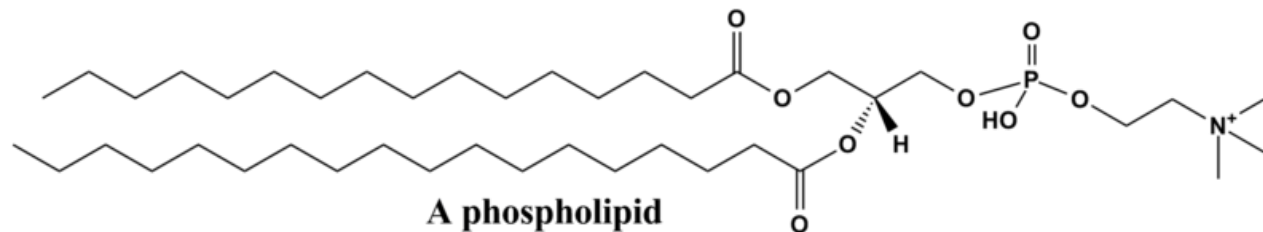
Cholesterol



A free fatty acid



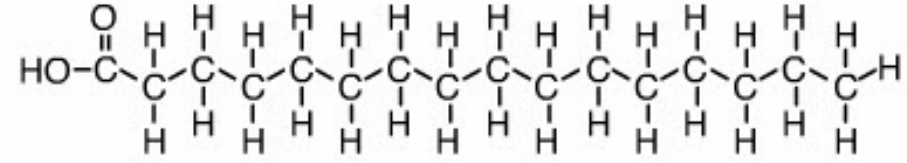
A triglyceride



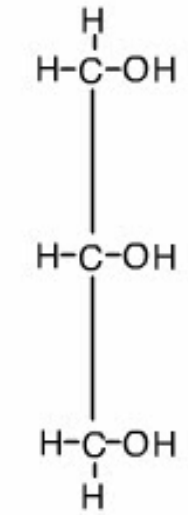
A phospholipid

Fats

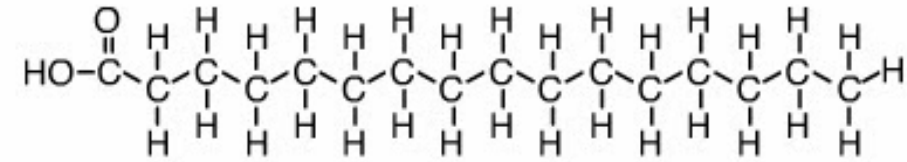
- or **triglyceride** (甘油三酯)
- Consists of a **glycerol** (甘油) molecule joined with **three fatty acid** (脂肪酸) molecules via a dehydration reaction
- Essential functions in the human body
 - Energy storage
 - Cushioning
 - Insulation (隔热, 保温)



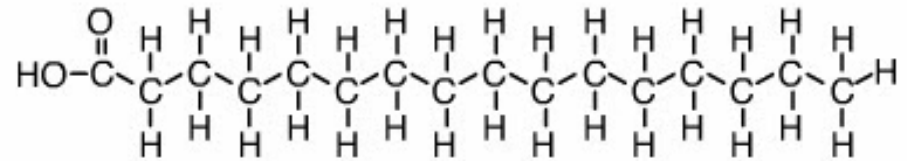
Fatty acid



Glycerol

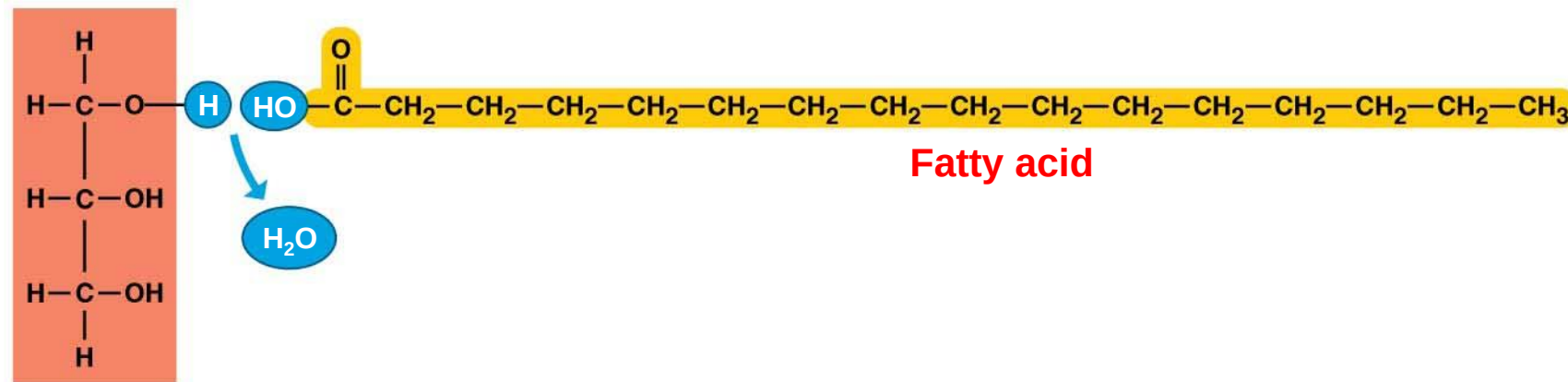


Fatty acid



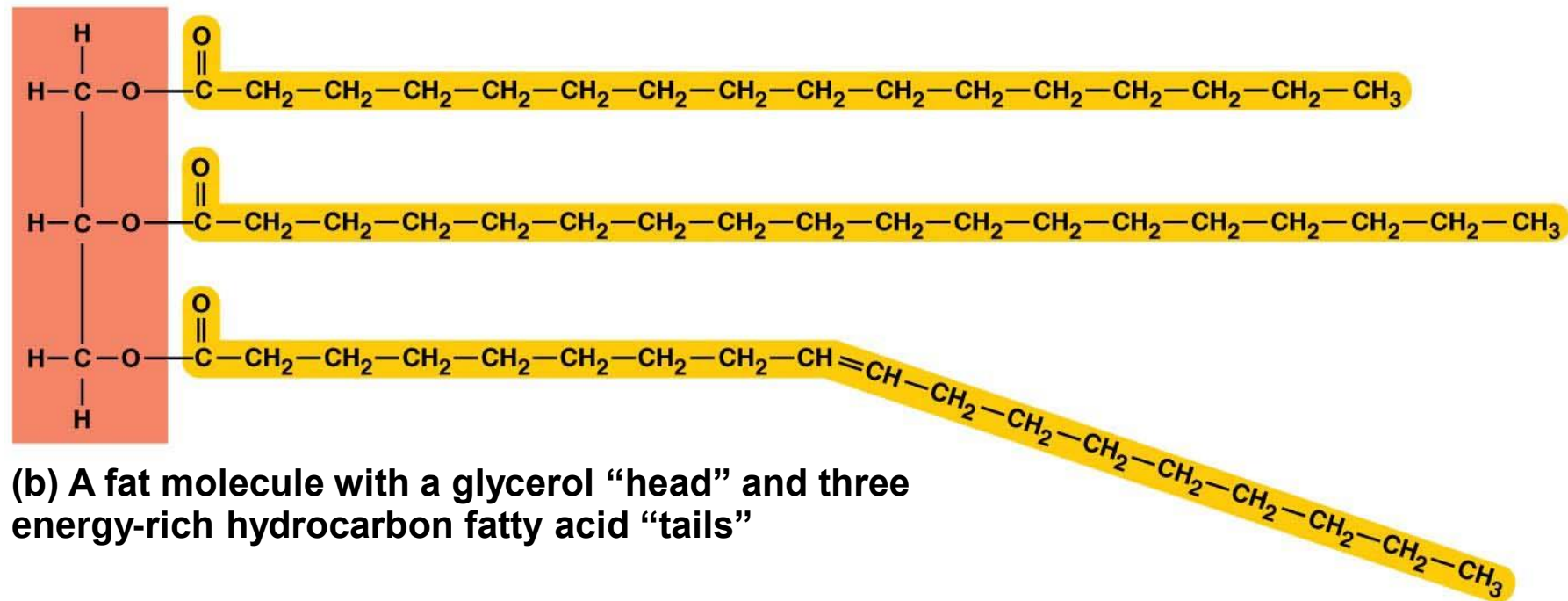
Fatty acid

Figure 3.11



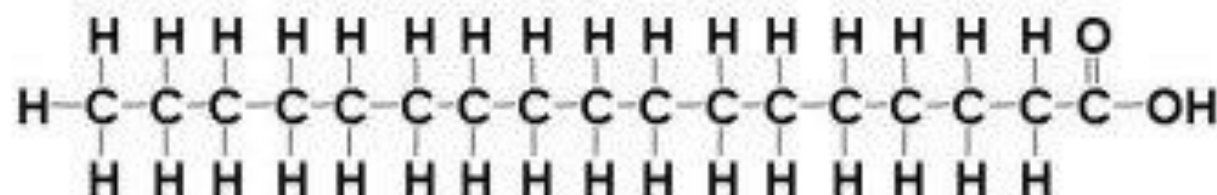
Glycerol (甘油)

(a) A dehydration reaction linking a fatty acid to glycerol

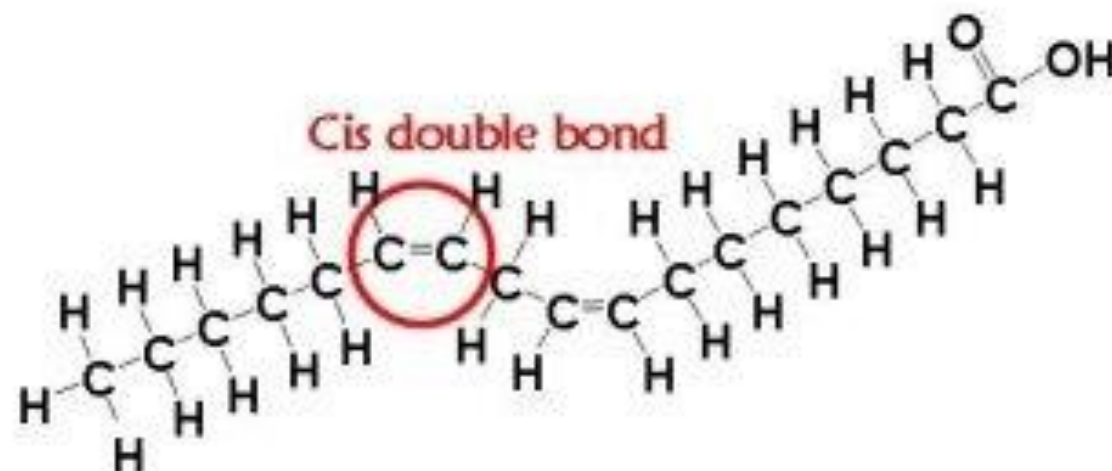


(b) A fat molecule with a glycerol “head” and three energy-rich hydrocarbon fatty acid “tails”

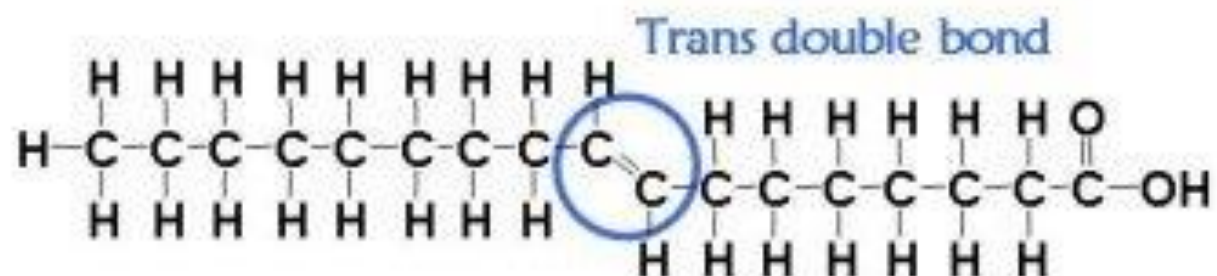
SATURATED
Stearic acid
(found in butter)



UNSATURATED
Linoleic acid
(found in vegetable oil)

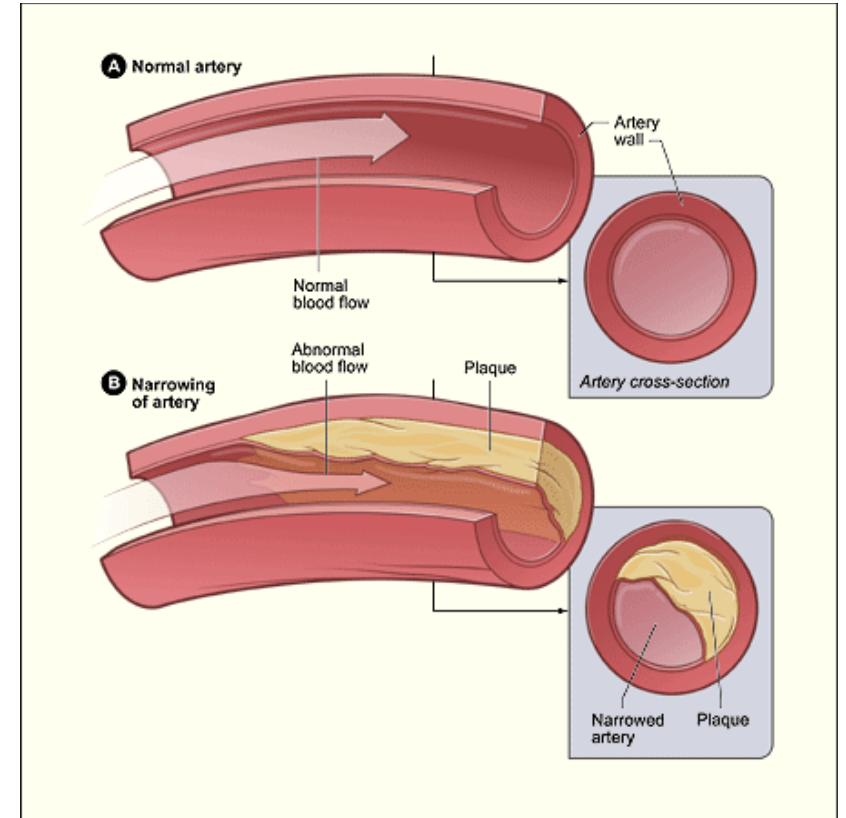


TRANS
trans-Linoleic acid
(found in some
margarine)



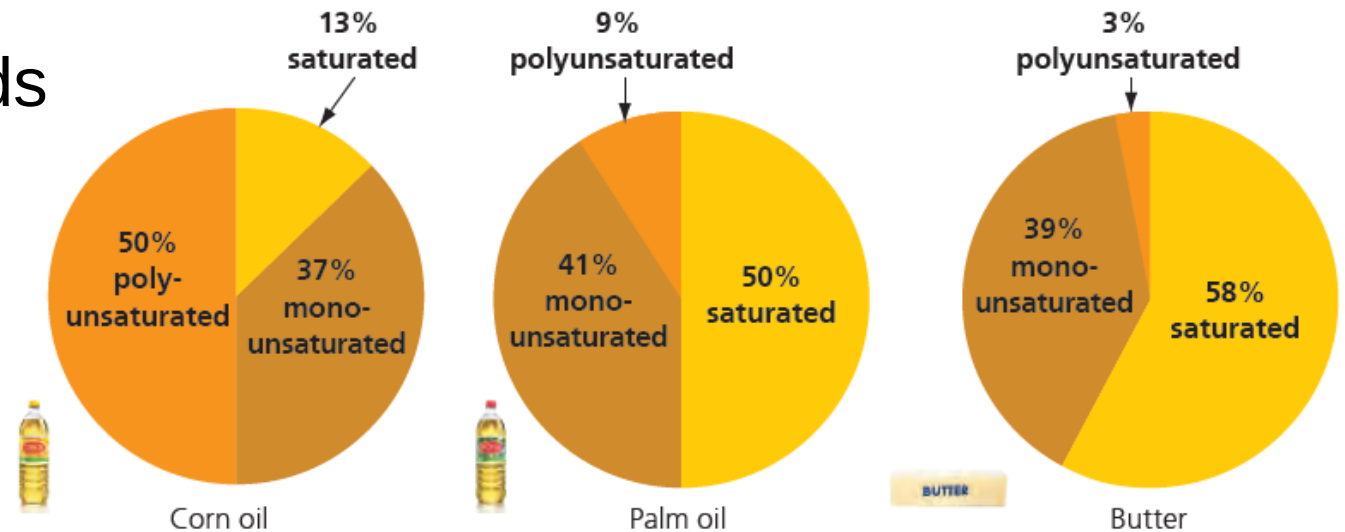
Saturated (饱和的) Fats

- **Saturated**, if it has the **maximum** number of hydrogens
 - A saturated fat has **all three** of its fatty acids saturated
- Most animal fats
 - A relatively **high proportion** of saturated fatty acids
 - Easily stack, tending to be **solid** at room temperature
 - Contribute to **atherosclerosis** (动脉粥样硬化), in which lipid-containing **plaques** (斑块) build up along the inside walls of blood vessels (血管)



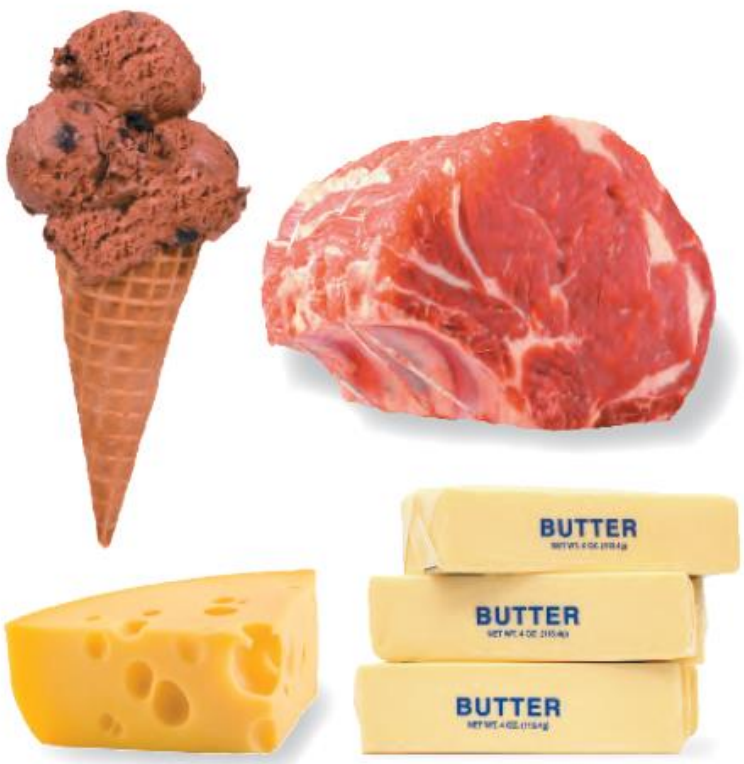
Unsaturated (不饱和的) Fats

- **Unsaturated**, if the carbon skeleton of a fatty acid has **fewer than the maximum** number of hydrogens at the double bond
- Monounsaturated
- Polyunsaturated
- Most plant and fish fats
 - high in unsaturated fatty acids
 - **liquid** at room temperature



Are all Unsaturated Fats Good for Health?

▼ **Figure 3.13** Types of fats.

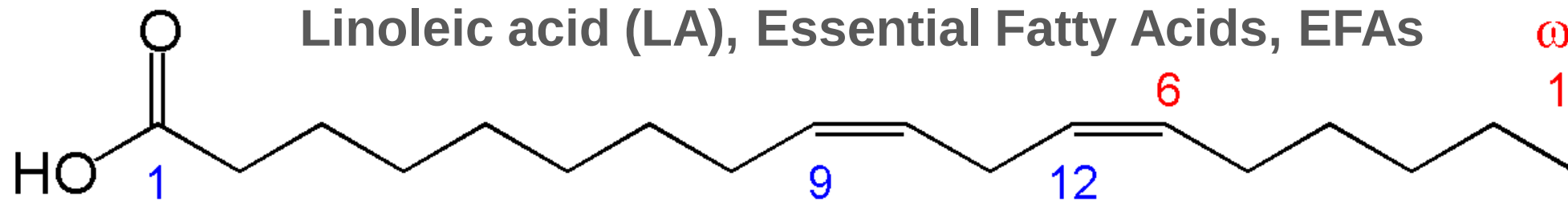
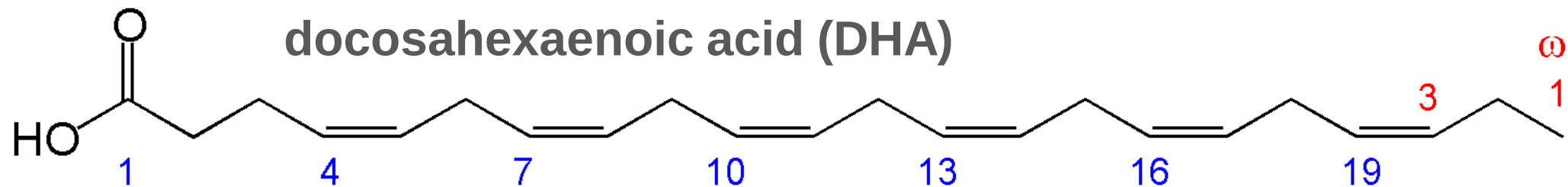
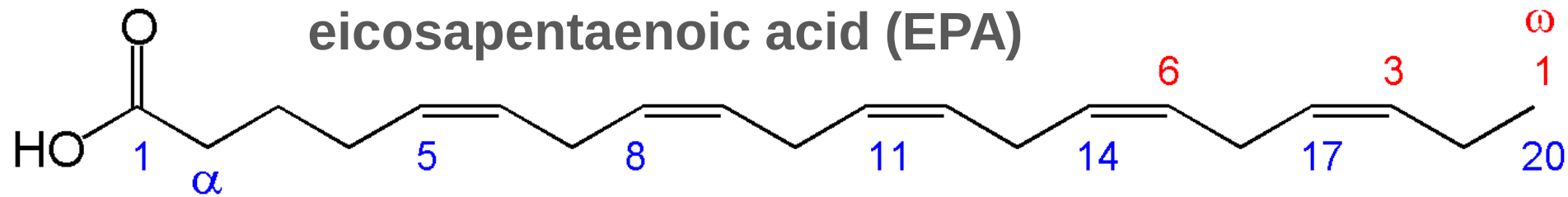
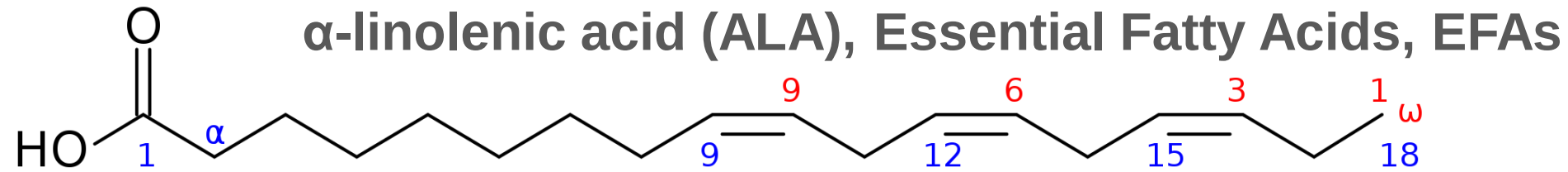
Saturated Fats (unhealthy fats found primarily in meat and full-fat dairy products; solid at room temperature)	Unsaturated Fats (fats found primarily in fish and plants; usually liquid at room temperature)
	 Plant and fish oils (unhydrogenated; usually liquid at room temperature)  INGREDIENTS: SOYBEAN OIL, FULLY HYDROGENATED COTTONSEED OIL, PARTIALLY HYDROGENATED COTTONSEED AND SOYBEAN OILS, MONO AND DIGLYCERIDES, TBHQ AND CITRIC ACID (ANTIOXIDANTS). Trans fats (unhealthy fats in hydrogenated processed foods; solid at room temperature)    Omega-3 fats (beneficial fats in some fish and plant oils; liquid at room temperature)

(反式脂肪)

Hydrogenation and Trans Fats

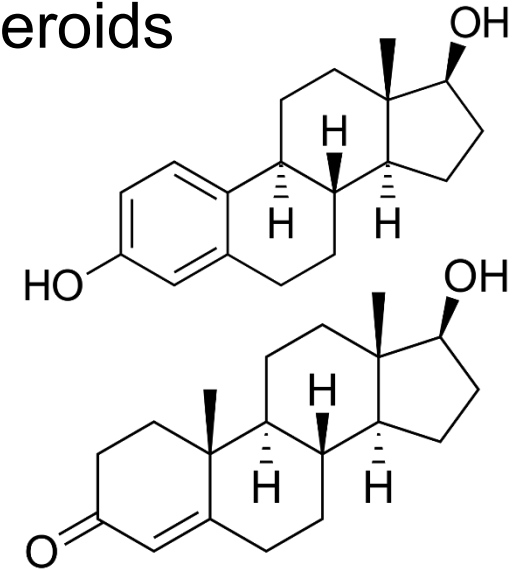
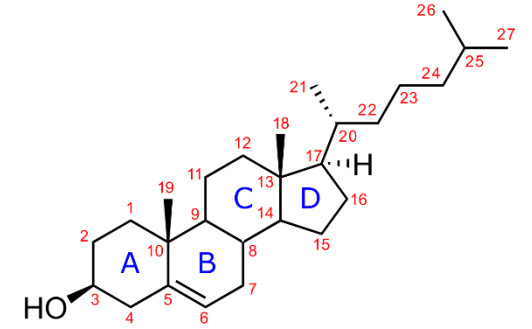
- “adds hydrogen”
- Converts unsaturated fats to saturated fats
 - Economic considerations
 - Stabilize the product
 - Extend the shelf life (保质期)
- Makes liquid fats solid at room temperature
- Creates **trans fatty acids (TFAs)**, a type of **unsaturated** fat that is **particularly bad** for your cardiovascular health

Omega-3 and Omega-6 fatty acids

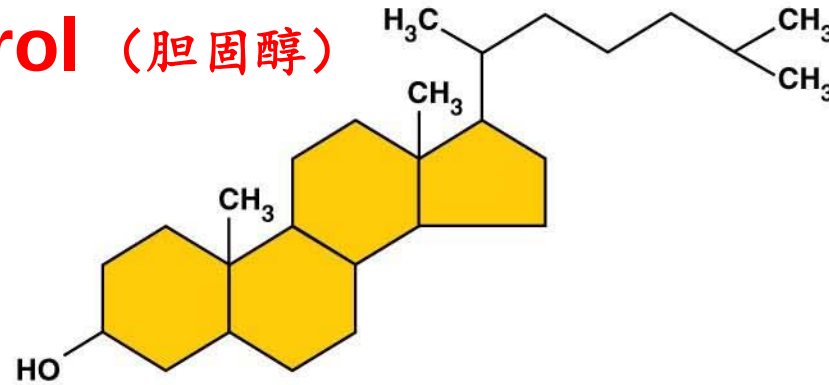


Steroids (类固醇)

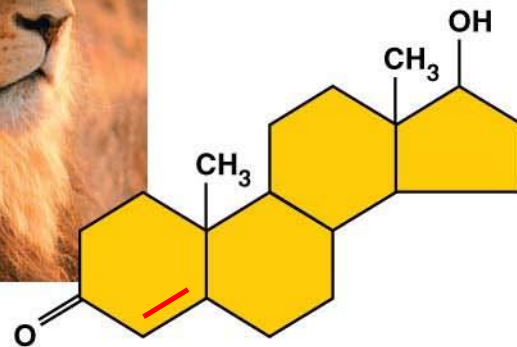
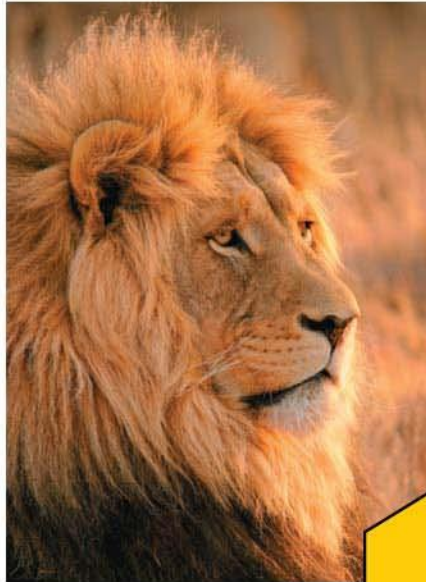
- Different from fats in structure and function
- Named after **Cholesterol** (胆固醇)
 - the “**base steroid**” from which your body produces other steroids
- Signaling molecules
 - Estradiol (雌二醇), an estrogen (雌激素)
 - Testosterone (睾酮)
- Important components of **cell membranes** that alter **membrane fluidity**



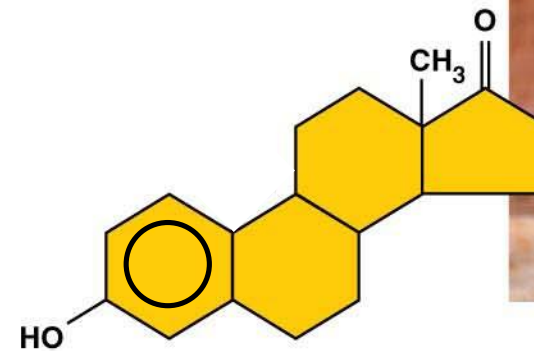
Cholesterol (胆固醇)



can be converted
by the body to



Testosterone



Estrone
(A type of estrogen)

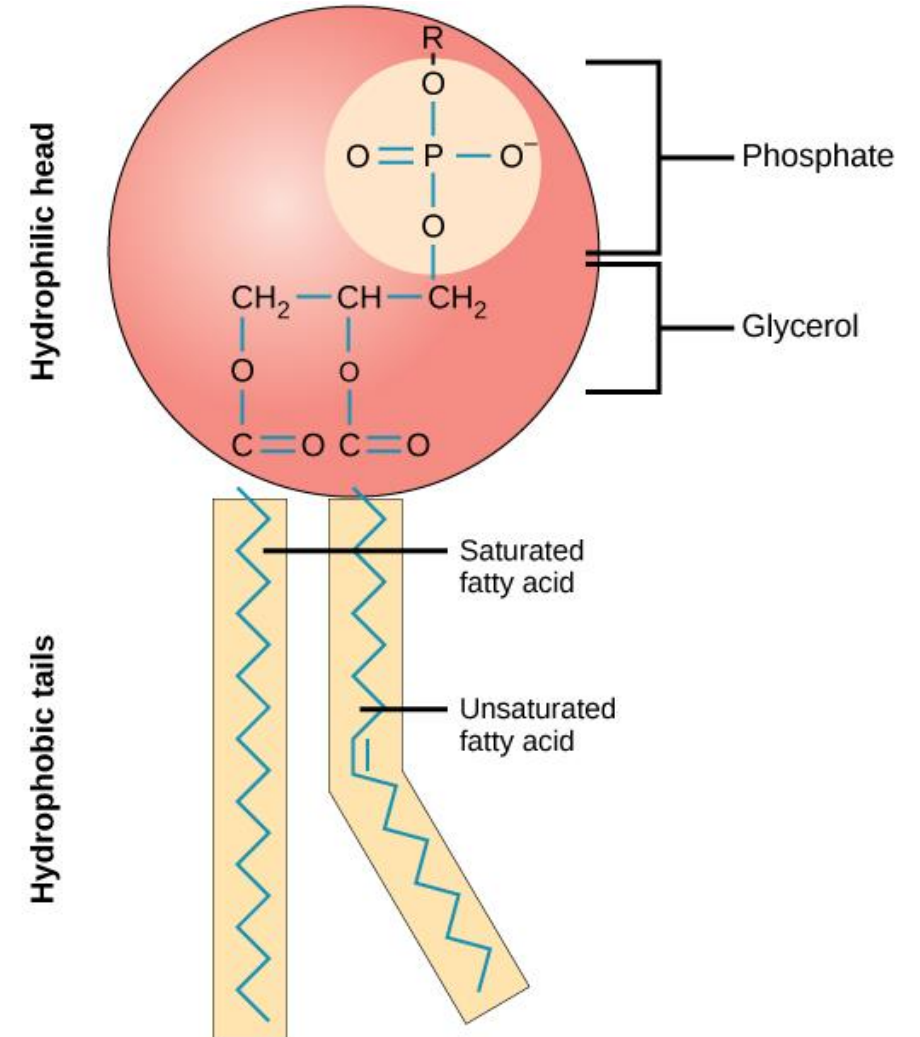
Synthetic anabolic steroids (合成类固醇)

- Variants of testosterone, mimic some of its effects
- Originally prescribed to treat diseases such as cancer and AIDS
- Abused by athletes to build up their muscles quickly
- Serious physical and mental problems
- **Most** athletic organizations ban the use of anabolic steroids (stimulant) because of their many potential health hazards coupled with the unfairness of an artificial advantage.



Phospholipids (磷脂)

- A **hydrophilic** “head” with a negatively charged phosphate group
- Two nonpolar **hydrophobic** fatty acid “tails”.
 - One of the “tails” contains **cis bond**
 - EPA and DHA in marine phospholipids

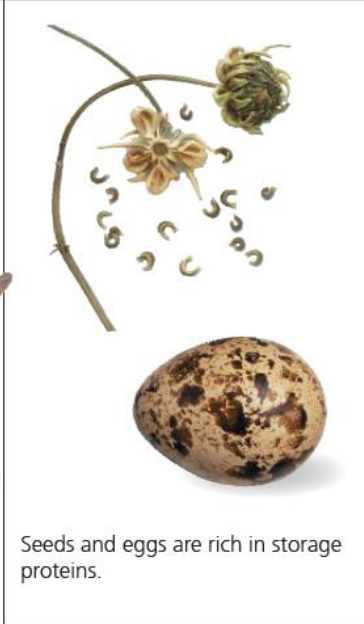
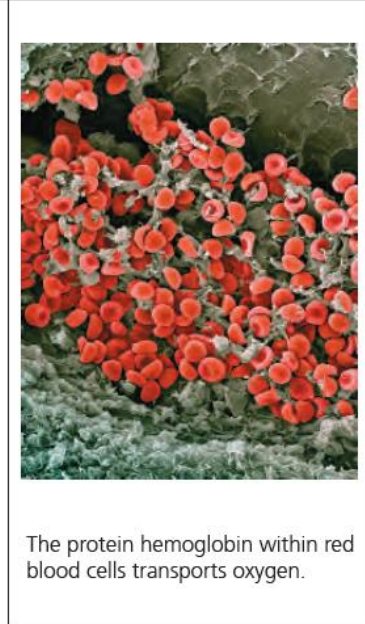


Proteins (蛋白质) are polymers of amino acid (氨基酸) monomers (单体)

LO: Describe the structure of proteins and explain how the structure of a protein relates to its function.

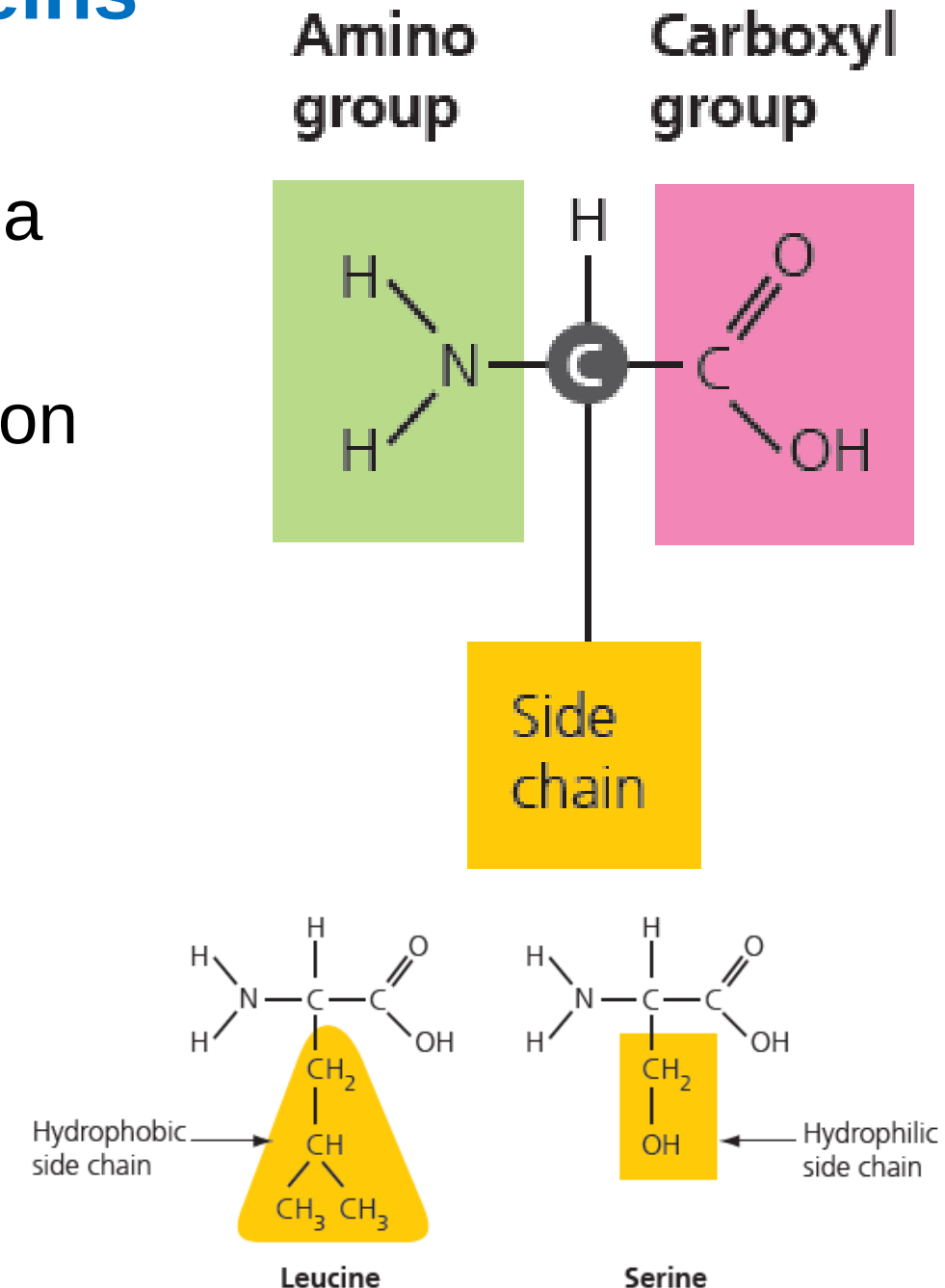
- more than 50% of the dry weight of most cells

▼ **Figure 3.16** Some of the varied roles played by proteins.

MAJOR TYPES OF PROTEINS				
Structural Proteins (provide support)	Storage Proteins (provide amino acids for growth)	Contractile Proteins (help movement)	Transport Proteins (help transport substances)	Enzymes (help chemical reactions)
 Structural proteins form hair, ligaments, and horns.	 Seeds and eggs are rich in storage proteins.	 Contractile proteins enable muscles to contract.	 The protein hemoglobin within red blood cells transports oxygen.	 Some cleaning products use enzymes to help break down molecules.

Amino Acids, the Monomers of Proteins

- All proteins are made by stringing together a **common set of 20** kinds of amino acids.
- Every amino acid consists of a central carbon atom bonded to four covalent partners:
 - **Three common** groups to all amino acids:
 - a carboxyl group ($-\text{COOH}$)
 - an amino group ($-\text{NH}_2$)
 - a hydrogen atom
 - **One unique** side chain



TWENTY-ONE PROTEINOGENIC α -AMINO ACIDS

Side chain charge
at physiological
pH 7.4

pK_a values shown
italicized

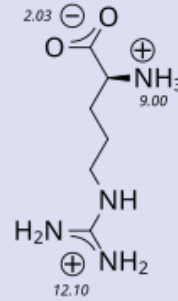
⊕ Positive
⊖ Negative

A. Amino Acids with Electrically Charged Side Chains

Positive

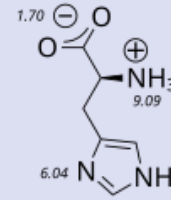
Arginine

Arg R



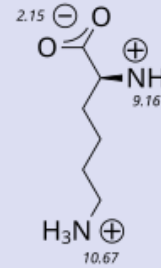
Histidine

His H



Lysine

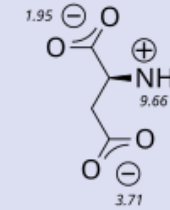
Lys K



Negative

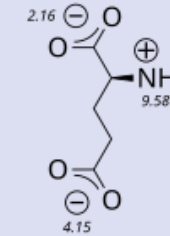
Aspartic Acid

Asp D



Glutamic Acid

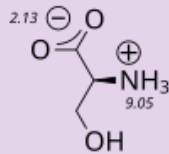
Glu E



B. Amino Acids with Polar Uncharged Side Chains

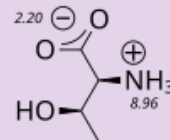
Serine

Ser S



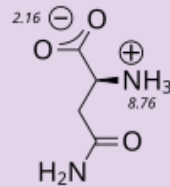
Threonine

Thr T



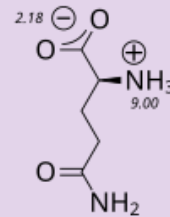
Asparagine

Asn N



Glutamine

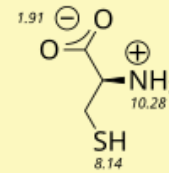
Gln Q



C. Special Cases

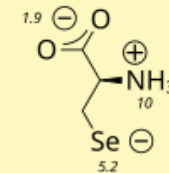
Cysteine

Cys C



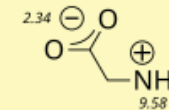
Selenocysteine

Sec U



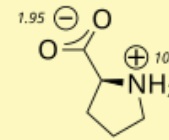
Glycine

Gly G



Proline

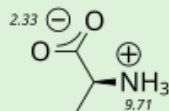
Pro P



D. Amino Acids with Hydrophobic Side Chains

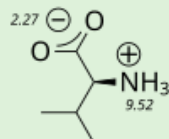
Alanine

Ala A



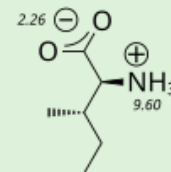
Valine

Val V



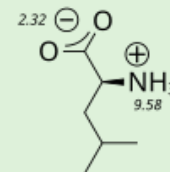
Isoleucine

Ile I



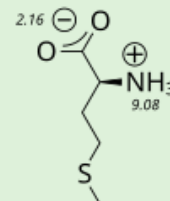
Leucine

Leu L



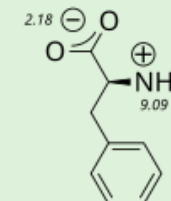
Methionine

Met M



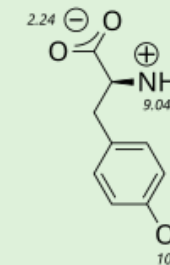
Phenylalanine

Phe F



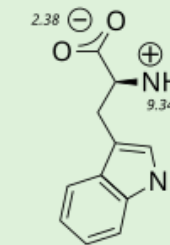
Tyrosine

Tyr Y



Tryptophan

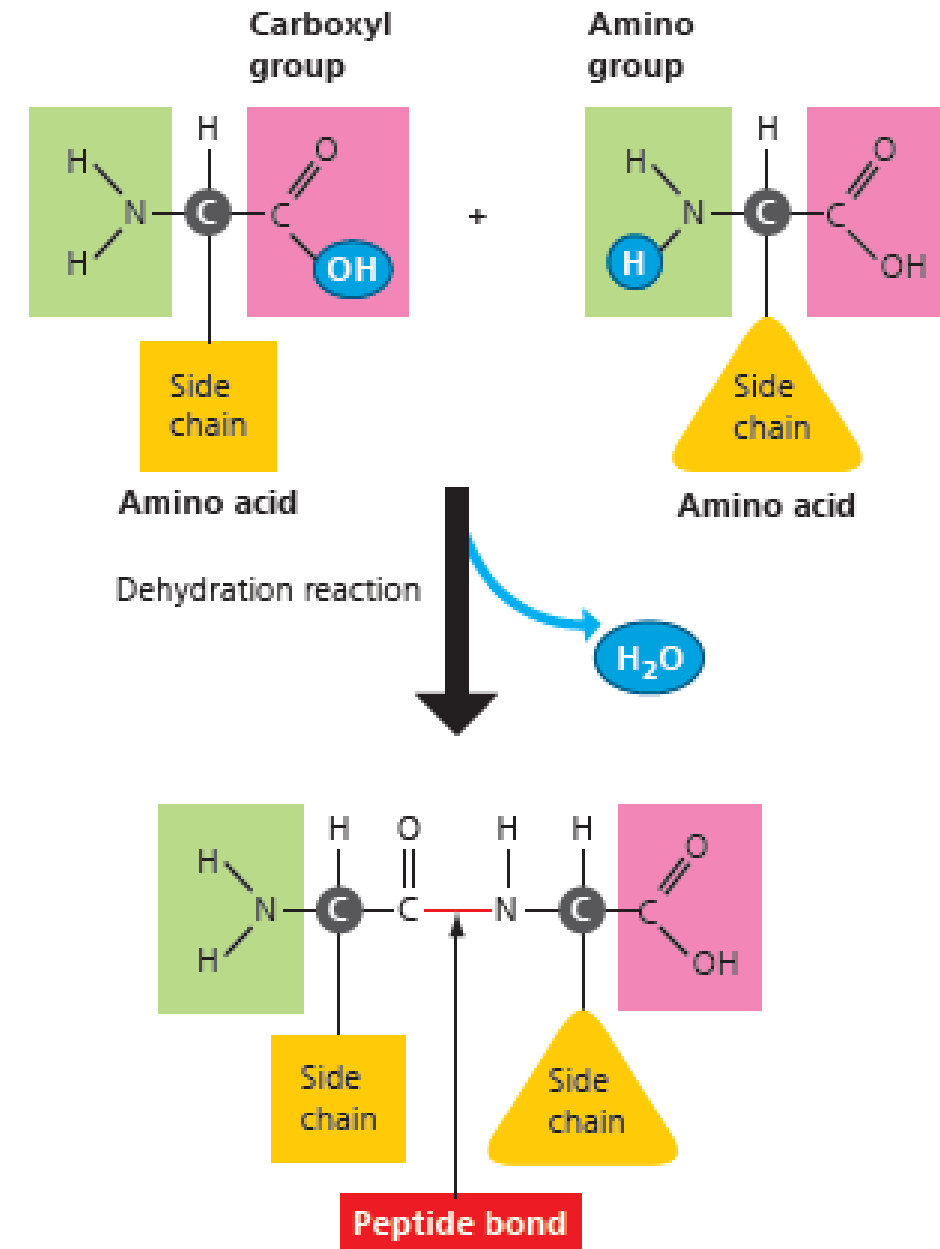
Trp W



Peptide Bonds (肽键)

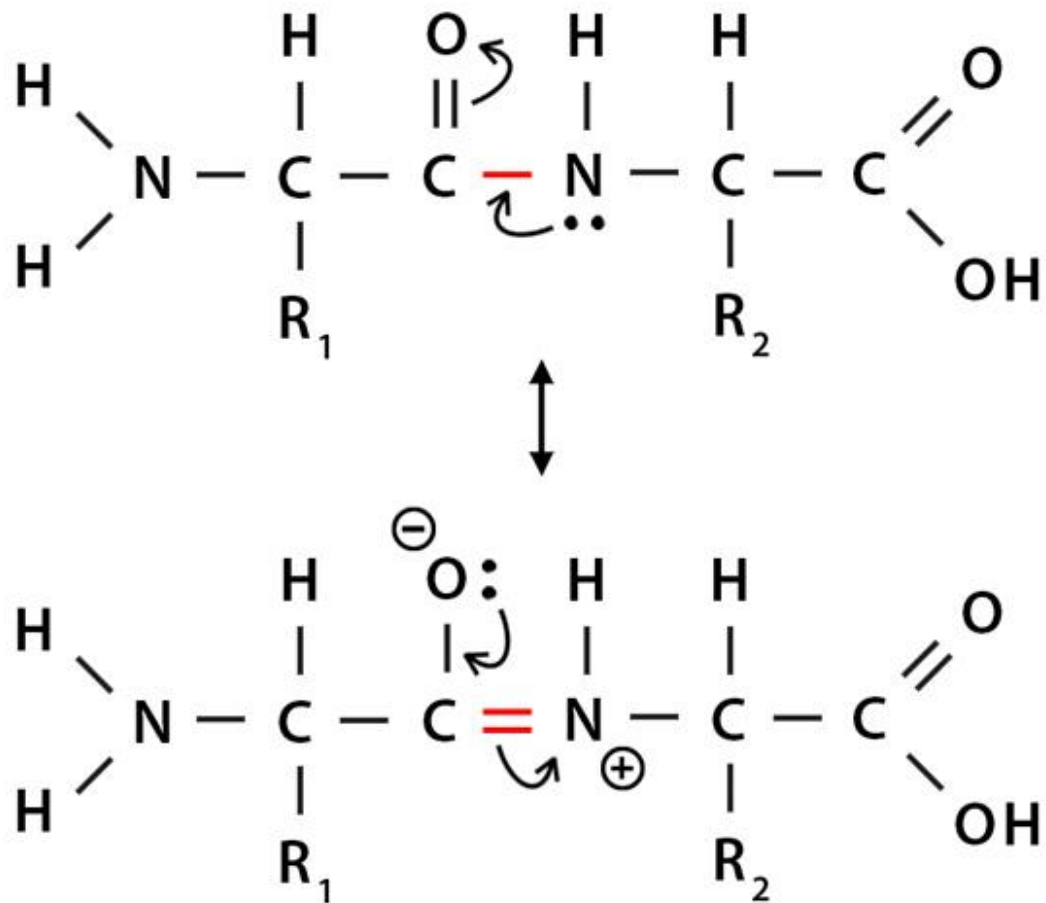
- Cells (ribosomes) link amino acids together by dehydration reactions.
 - forming peptide bonds (肽键)
 - creating long chains of amino acids called **polypeptides** (多肽)
- A functional protein is one or more polypeptide chains precisely twisted, folded, and coiled into a molecule of unique shape.

▼ **Figure 3.18** Joining amino acids. A dehydration reaction links adjacent amino acids by a peptide bond.

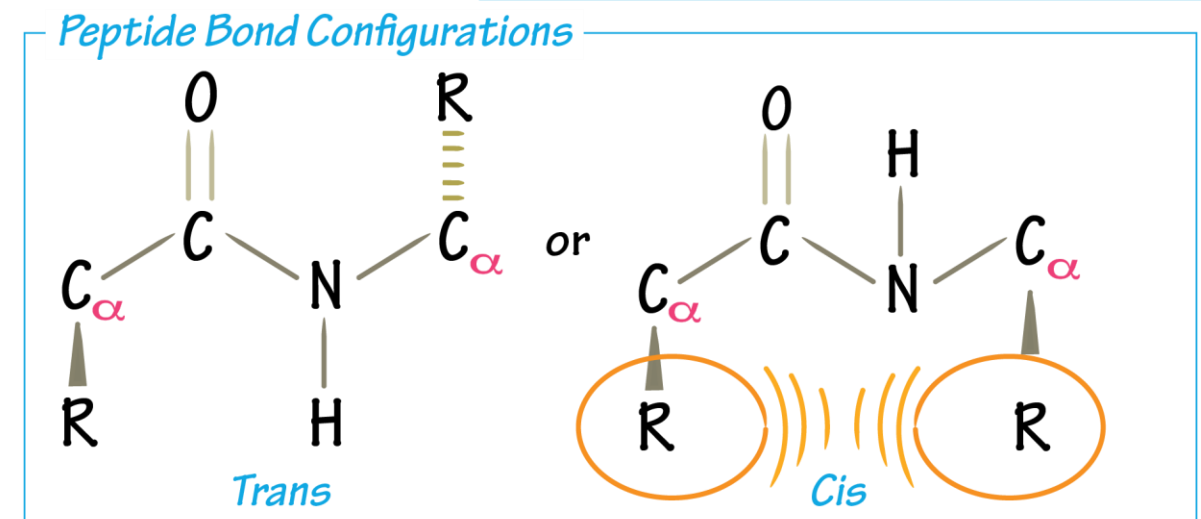
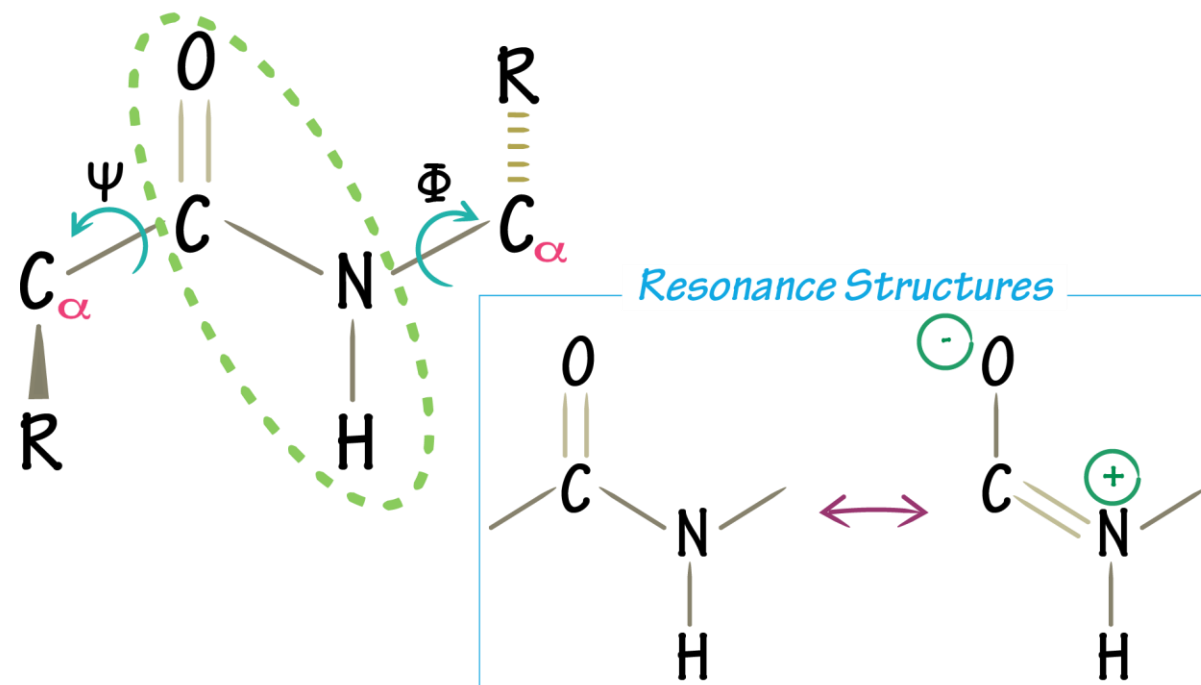


Peptide Bond Resonance Structure

Dipeptide



ChemistryLearner.com

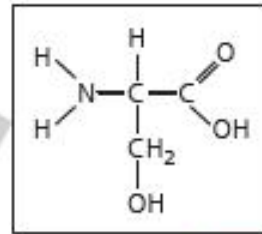


Protein Structure and Function

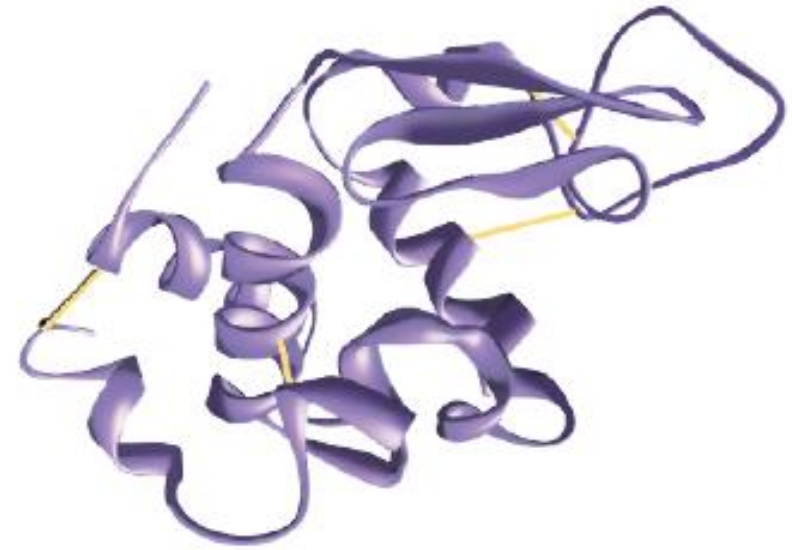
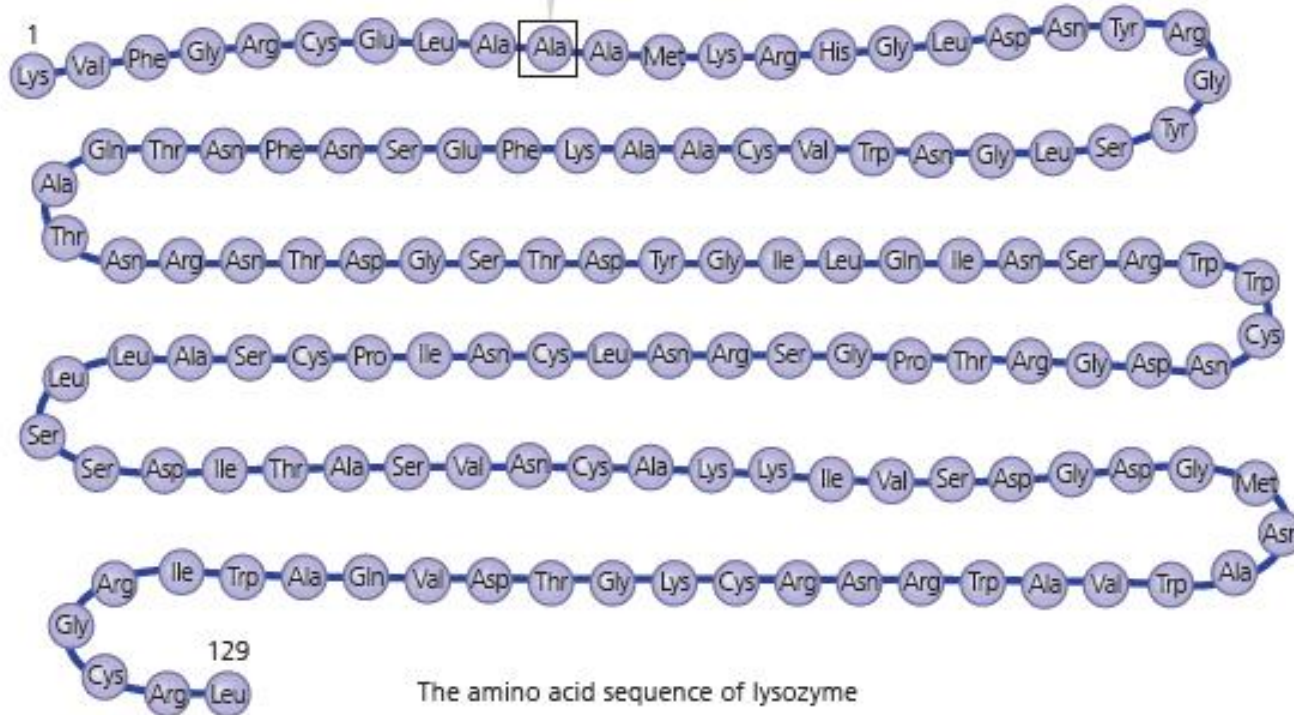
- Proteins use 20 different “letters” (amino acids) to create polypeptides hundreds or thousands of amino acids in length, to make the huge variety of proteins.
- The amino acid **sequence** of each polypeptide **determines** the three-dimensional **structure** of the protein.
- A protein’s three-dimensional **structure** enables the molecule to carry out its specific **function**.
- Nearly all proteins work by **recognizing and binding** to some other molecule.

snake that allows it to do its job.

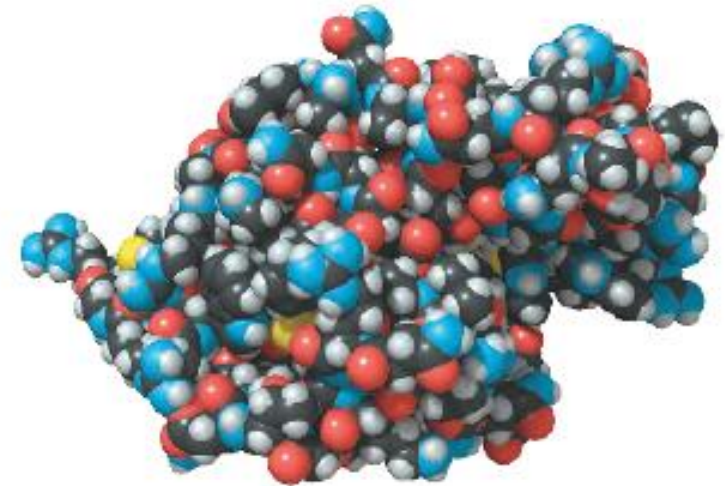
▼ **Figure 3.19 The structure of a protein.** The chain below, drawn in serpentine fashion so that it fits on the page, shows the amino acid sequence of a polypeptide found in lysozyme, an enzyme in your tears and sweat that helps prevent bacterial infections. The names of the amino acids are given as their three-letter abbreviations; for example, the amino acid alanine is abbreviated "Ala." This amino acid sequence folds into a protein of specific shape, shown in the two computer-generated representations. Without this specific shape, the protein could not perform its function.



One amino acid (alanine)



Here you can see how the polypeptide folds into a compact shape.



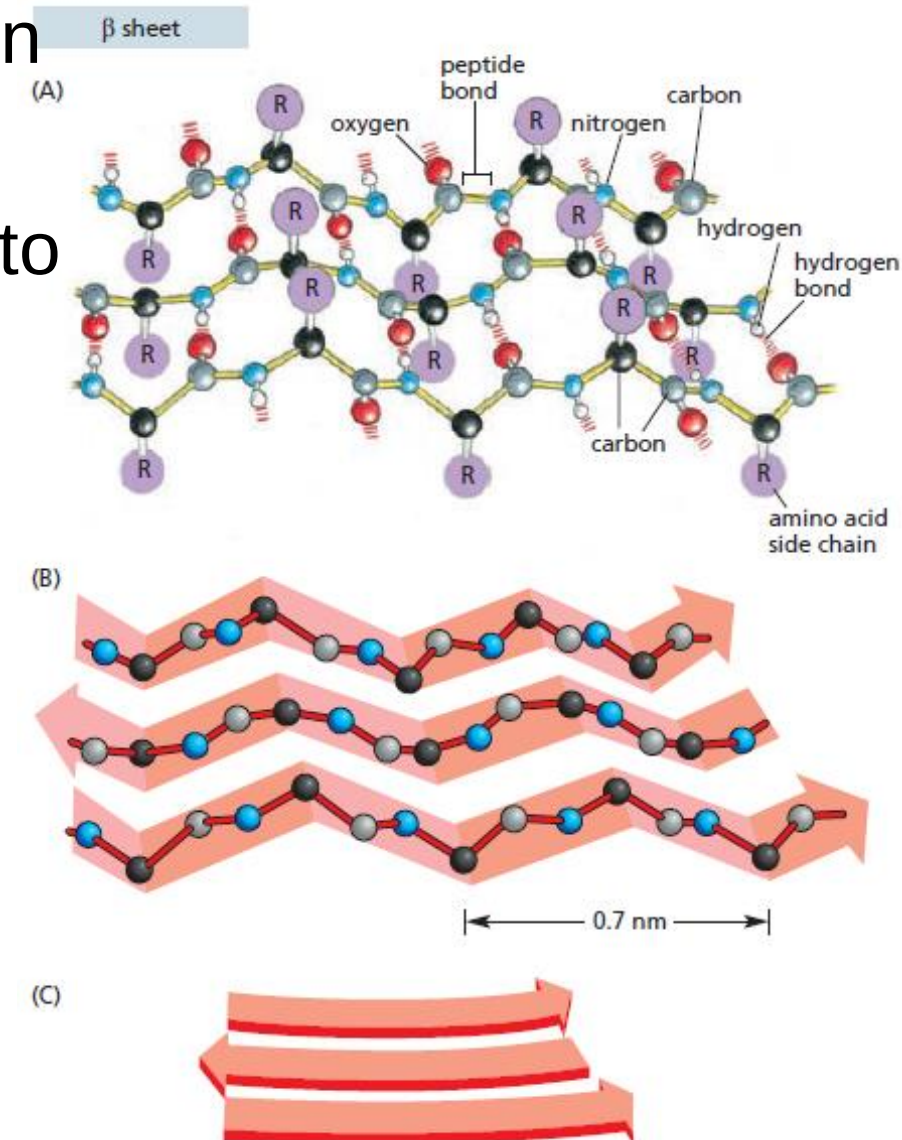
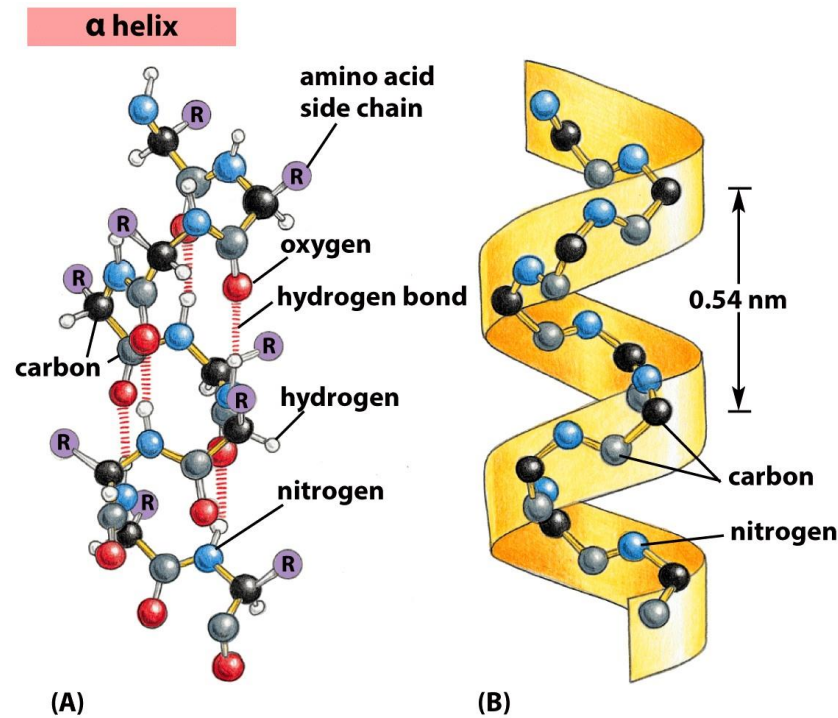
This model allows you to see the details of the protein's structure.

Viewing Proteins: SH2

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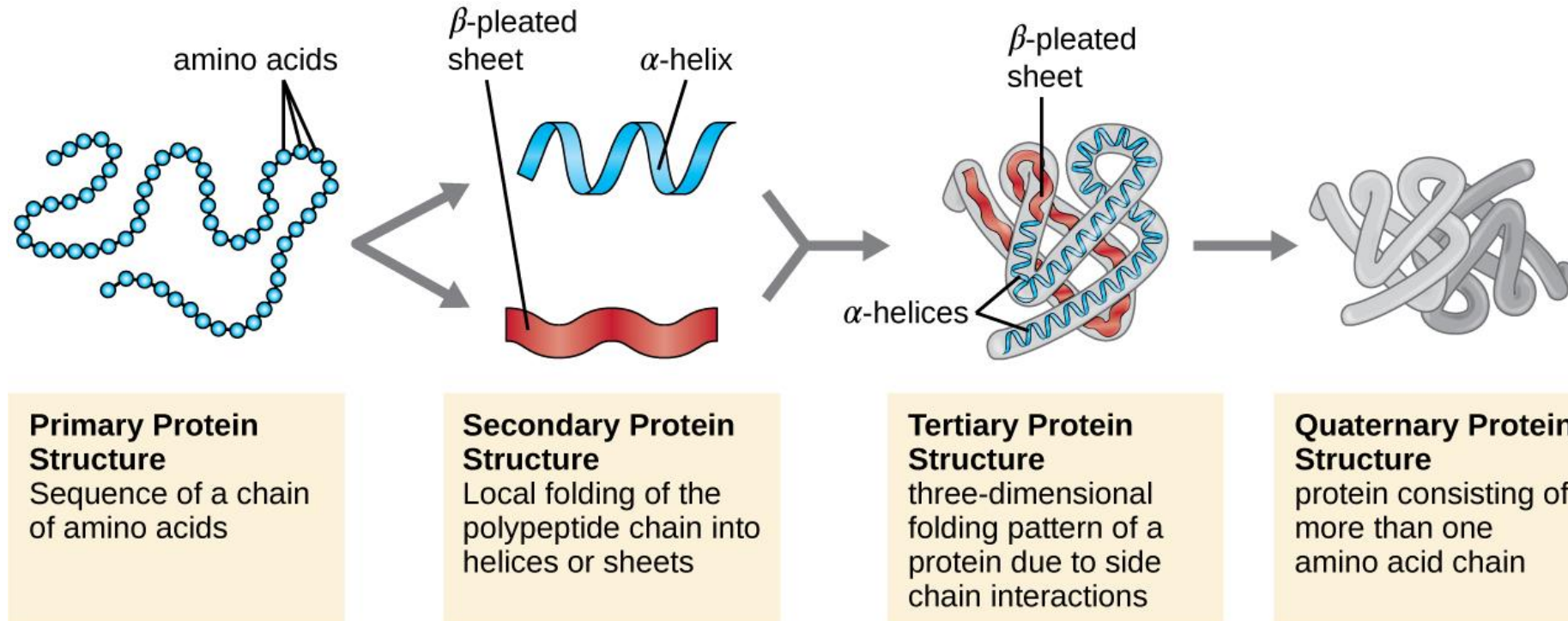
Four Level of Protein Structure 1

- **Primary Structure:** the **order** of amino acids in the peptide chain
- **Secondary Structure:** structures formed due to **hydrogen bonding** between **peptide bonds**



Four Level of Protein Structure 2

- **Tertiary** Structure (involves the three-dimensional folding of a protein due to interactions of amino acid **side chains**)
- **Quaternary** Structure (involves the association of **two or more polypeptides** into functional proteins)



All proteins Bind to Other Molecules

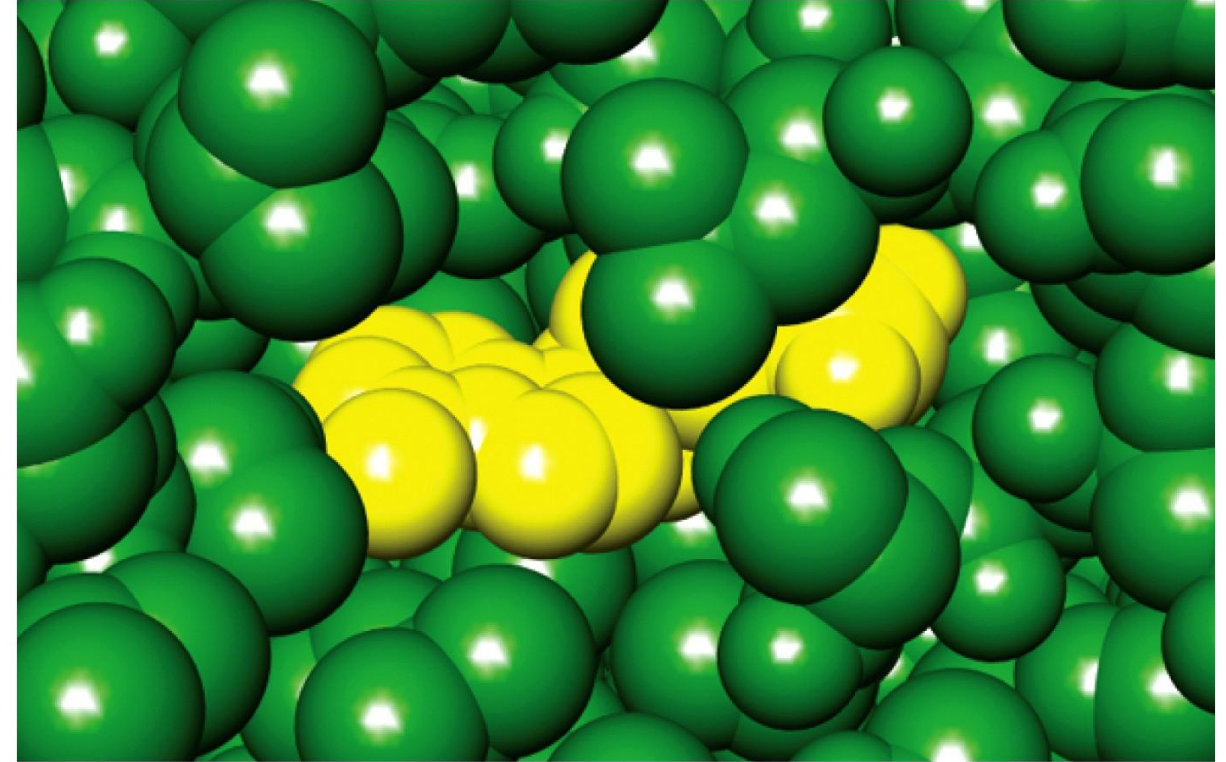
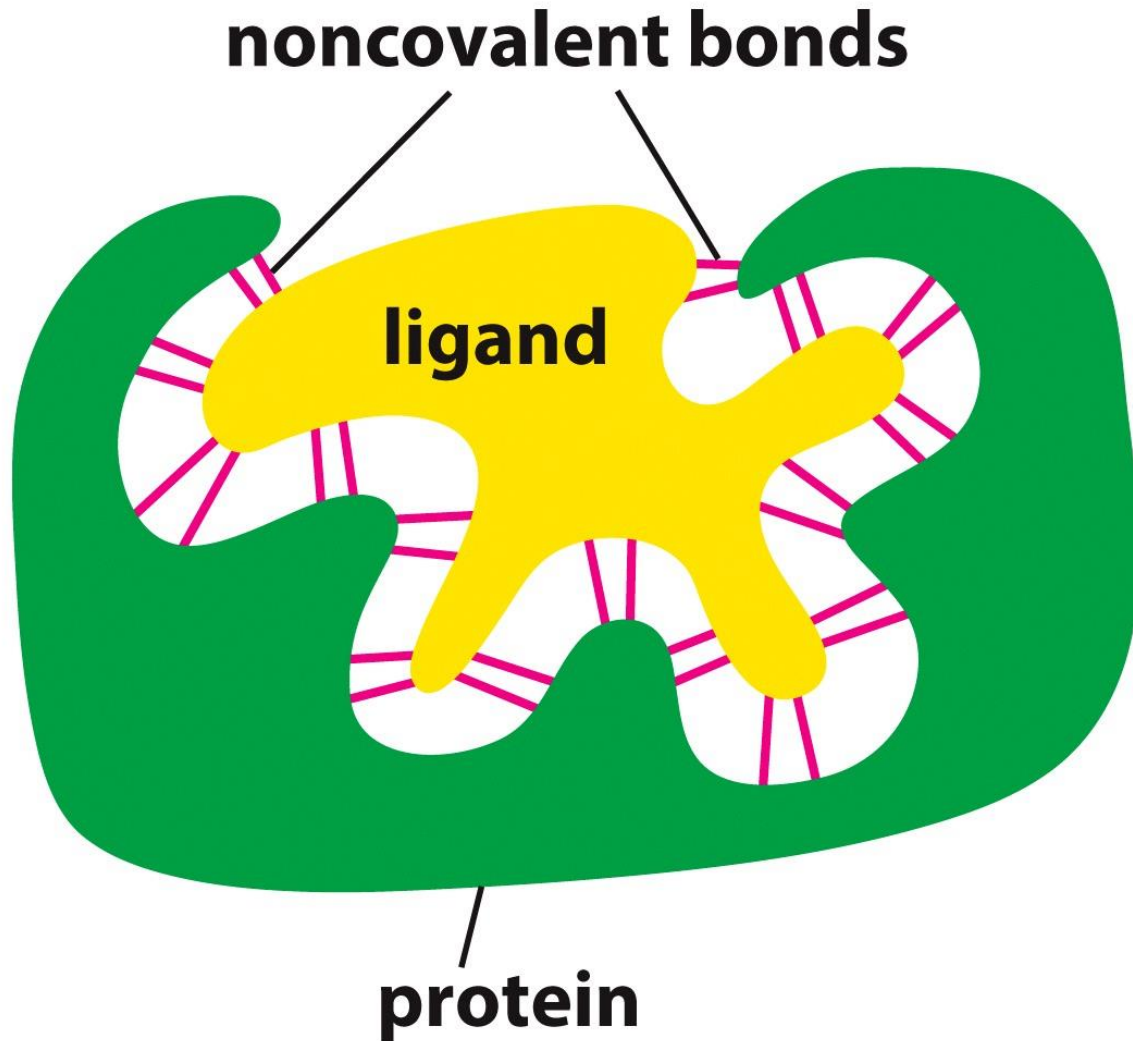


Figure 4-31b Essential Cell Biology, 4th ed. (© Garland Science 2014)

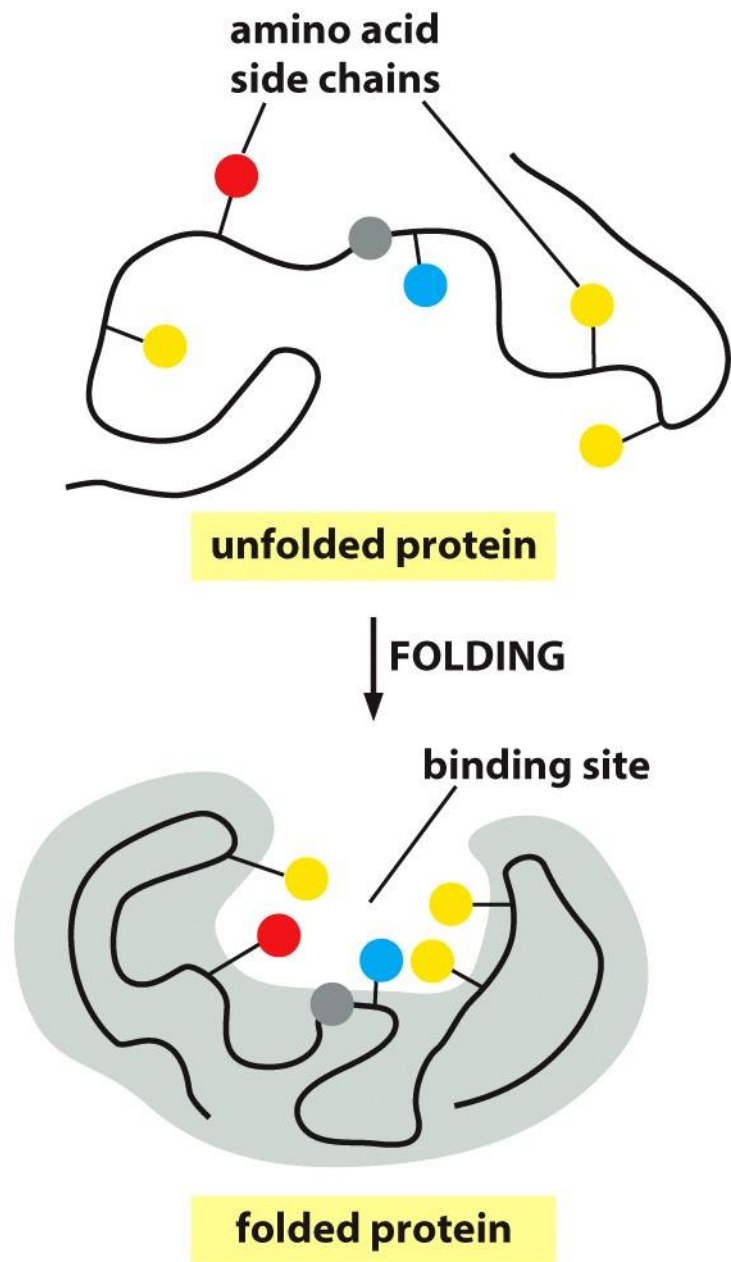


Figure 4-32a Essential Cell Biology, 4th ed. (© Garland Science 2014)

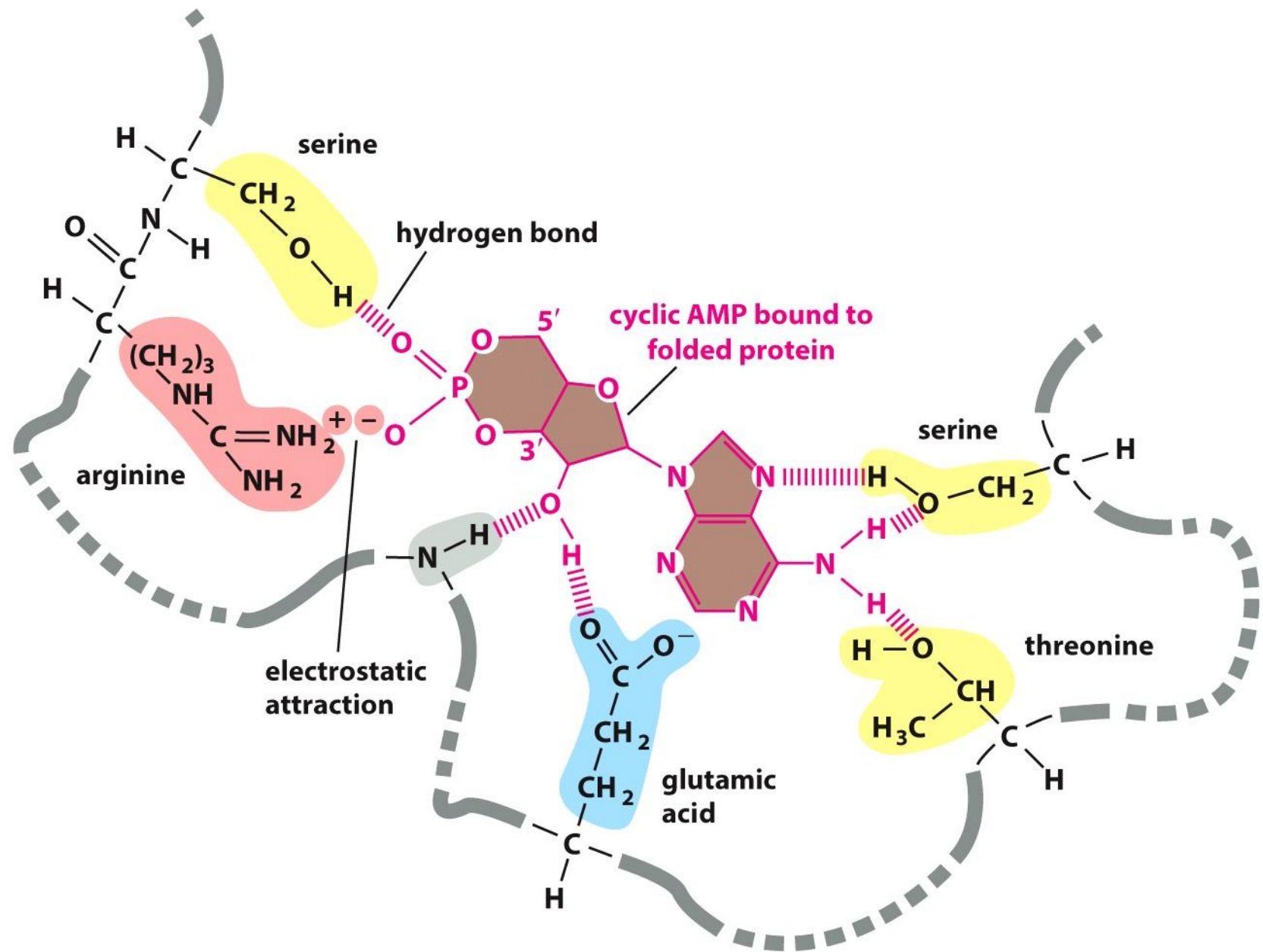
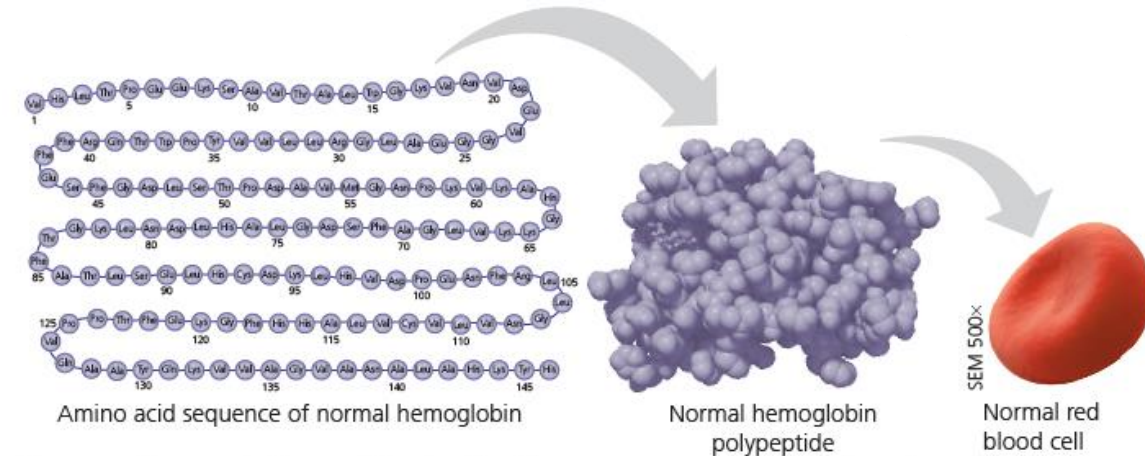


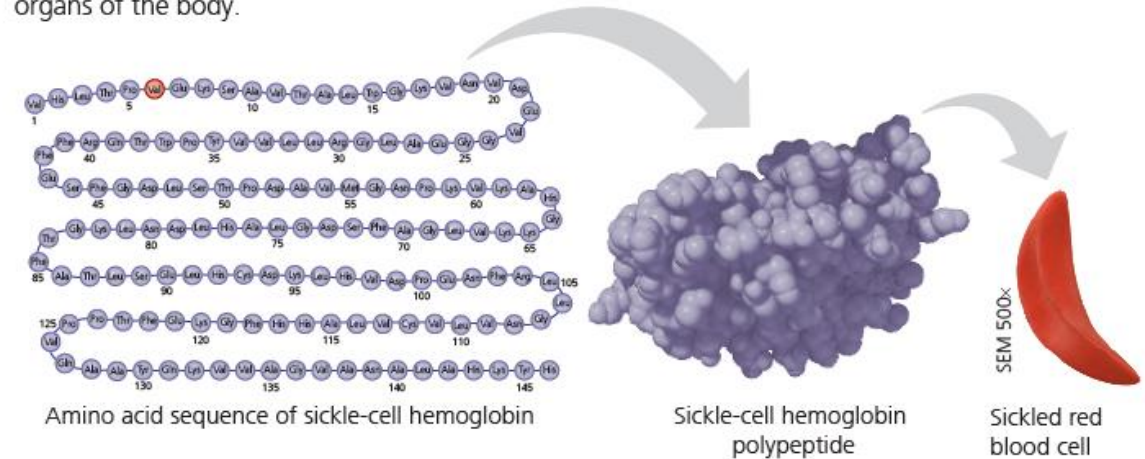
Figure 4-32b Essential Cell Biology, 4th ed. (© Garland Science 2014)

Protein Function

- A slight change in the amino acid sequence can affect a protein's ability to function.
- The substitution of **one amino acid** for another at a particular position in **hemoglobin** (血红蛋白), the blood protein that carries oxygen, causes **sickle-cell disease** (镰刀型细胞贫血症), an inherited blood disorder.



(a) Normal hemoglobin. Red blood cells of humans are normally disk-shaped. Each cell contains millions of molecules of the protein hemoglobin, which transports oxygen from the lungs to other organs of the body.



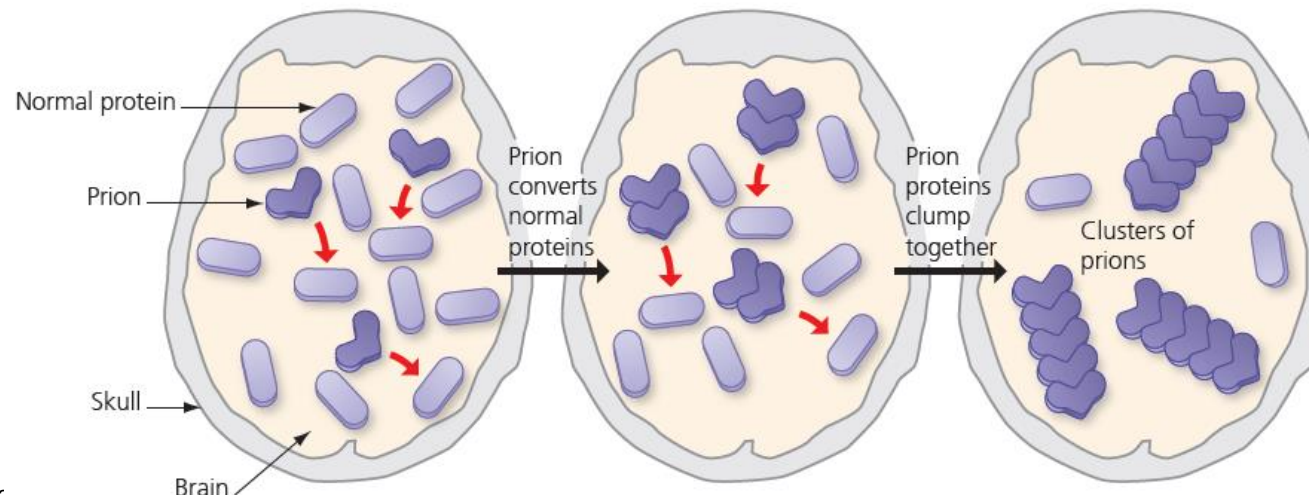
(b) Sickle-cell hemoglobin. A slight change in the amino acid sequence of hemoglobin causes sickle-cell disease. The inherited substitution of one amino acid—valine in place of the amino acid glutamic acid—occurs at the sixth position of the polymer. Such abnormal hemoglobin molecules tend to crystallize, deforming some cells into a sickle shape. Someone with the disease suffers from dangerous episodes when the angular cells clog tiny blood vessels, impeding blood flow.

Sickle Cell Anemia

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Prions(朊病毒), Misfolded Brain Proteins

- One difference in structure is enough to cause the protein to fold into a different shape, which alters its function, which in turn causes disease.
- Misfolded proteins are associated with many diseases, including some severe nervous system disorders.
- Prions are misfolded proteins with the ability to **transmit** their misfolded shape onto normal variants of the same protein



Prions cause fatal weight loss in several animals, including deer, elk, and moose. They also cause mad cow disease.

What other factors can cause a protein's shape change?

DS: Protein Design and Alpha Fold

- Baker Lab: www.bakerlab.org
- Foldit (<https://fold.it/science>), a computer game that enables you to contribute to cutting edge scientific research.
- AlphaFold Protein Structure Database: <https://alphafold.ebi.ac.uk>
- Use ColabFold (a faster, easier-to-run version of AlphaFold2):
 - Go to: <https://colabfold.com> (opens Google Colab notebooks)

Nobel Prize in Chemistry 2024



© Nobel Prize Outreach. Photo: Clément Morin
David Baker
Prize share: 1/2



© Nobel Prize Outreach. Photo: Clément Morin
Demis Hassabis
Prize share: 1/4



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John Jumper
Prize share: 1/4

The Nobel Prize in Chemistry 2024 was divided, one half awarded to David Baker "for computational protein design", the other half jointly to Demis Hassabis and John Jumper "for protein structure prediction"

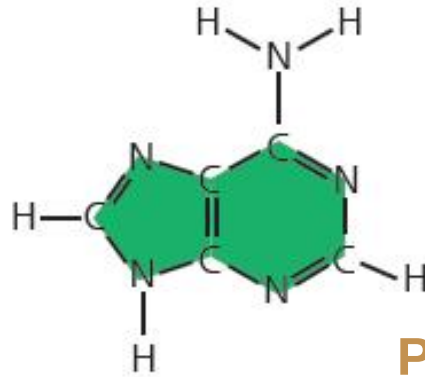
Nucleic acids (核酸)

LO:

- Review of knowledge of nucleic acids
- Identify the differences between DNA and RNA
- Macromolecules made from monomers nucleotides
 - store information
 - provide the instructions for building proteins
- Two types:
 - DNA (deoxyribonucleic acid)
 - RNA (ribonucleic acid)

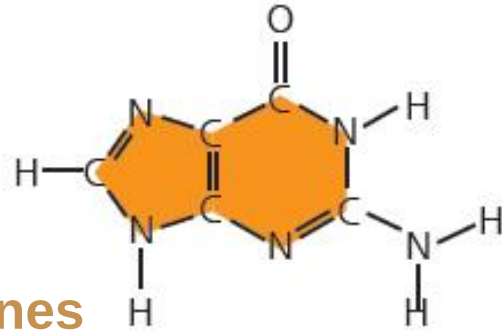
▼ **Figure 3.24** The nitrogenous bases of DNA.

Notice that adenine and guanine have double-ring structures. Thymine and cytosine have single-ring structures.

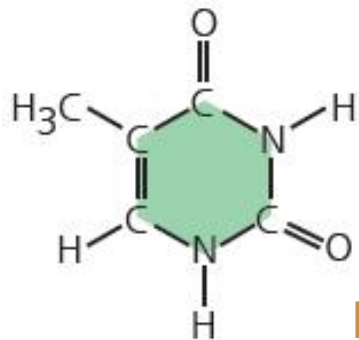


Adenine (A)

Purines

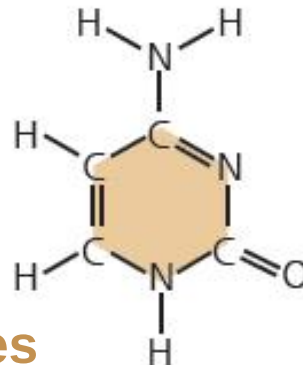


Guanine (G)

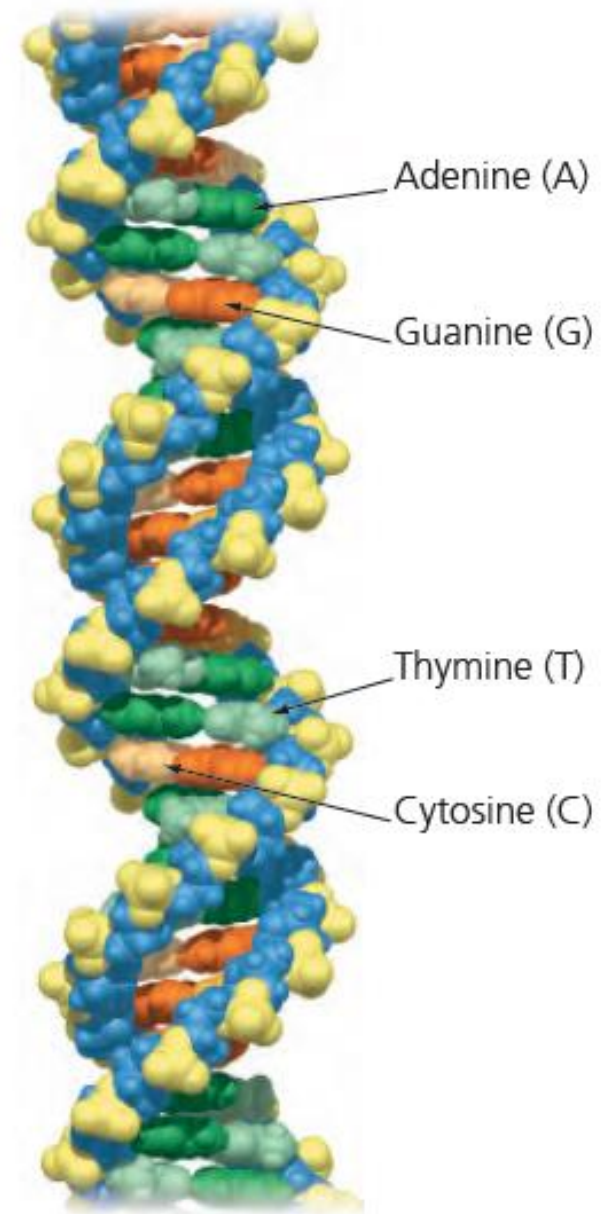


Thymine (T)

Pyrimidines



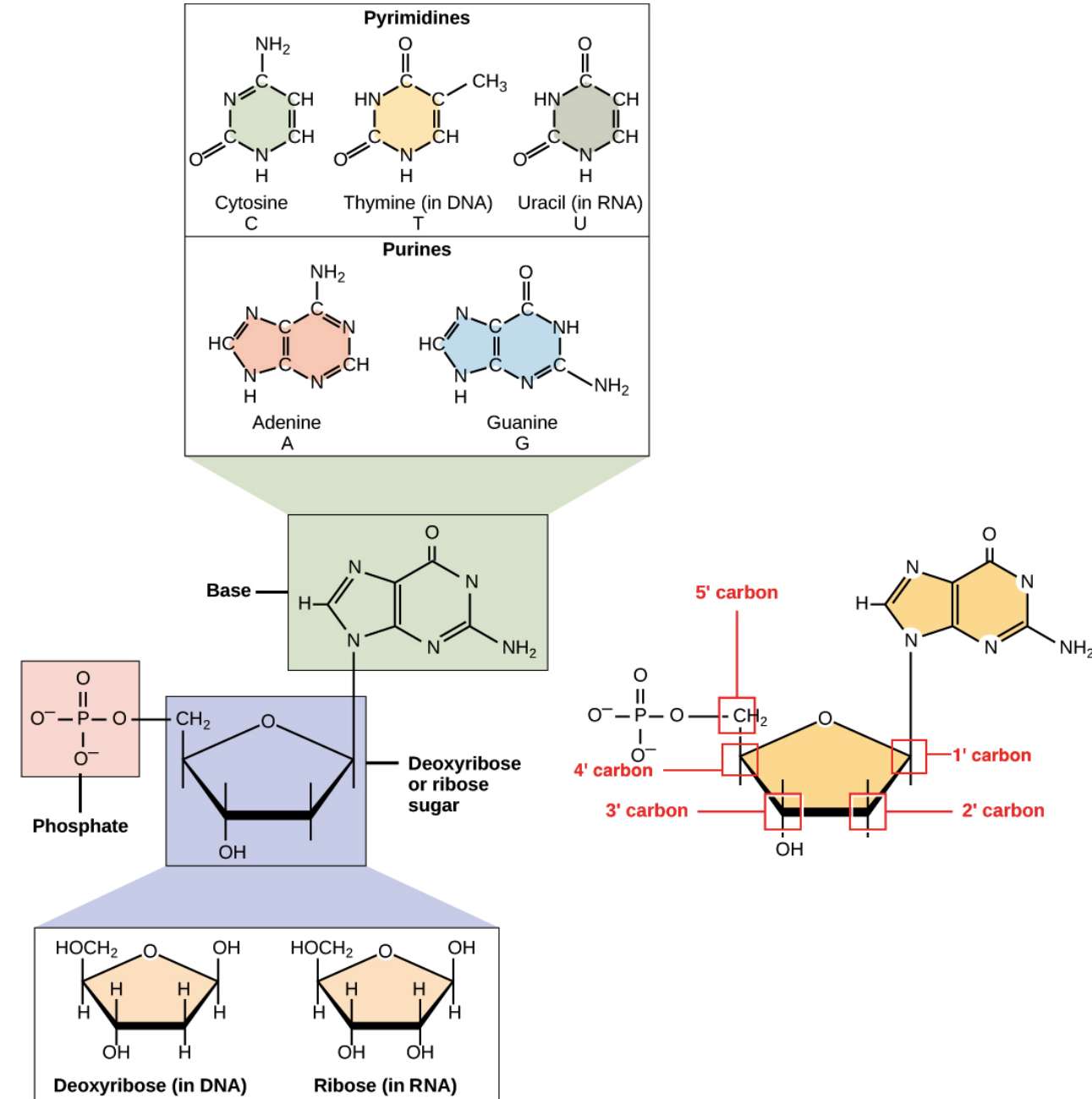
Cytosine (C)



Space-filling model of DNA
(showing the four bases in four different colors)

Differences between DNA and RNA

- Ribose (核糖) in RNA, Deoxyribose (脱氧核糖) in DNA
- Instead of the base thymine (T), RNA has uracil (U).
- RNA is usually found in single-stranded form, whereas DNA usually exists as a double helix.



Lactose Intolerance

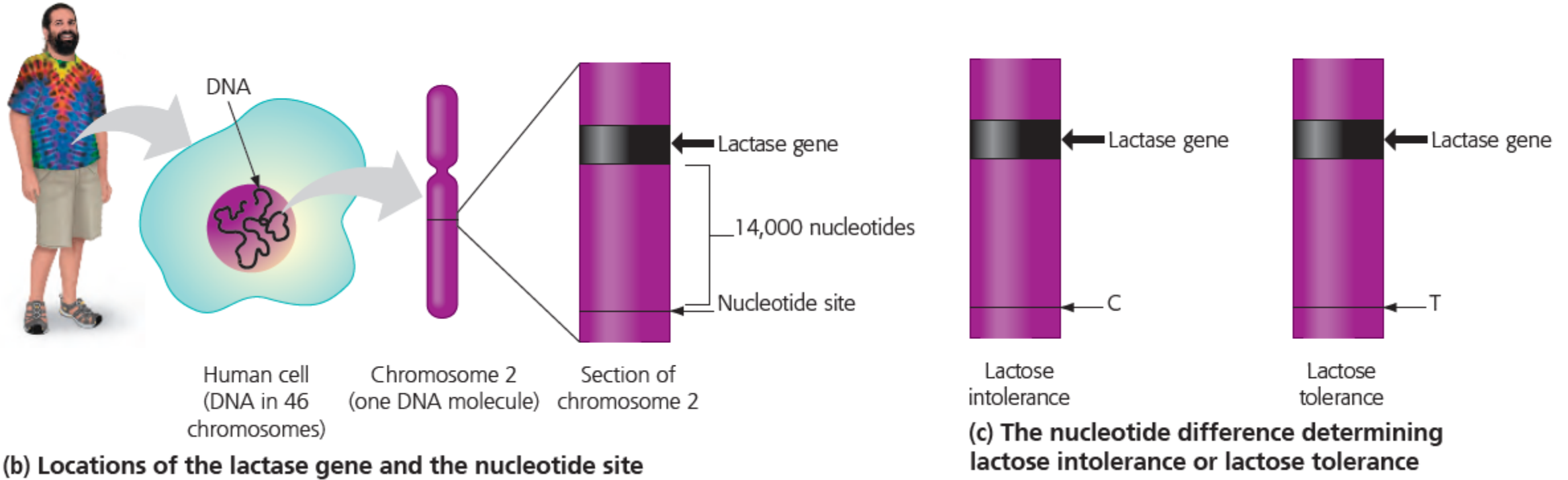
LO:

- Describe the causes and consequences of lactose intolerance.
- Explain why we think that lactose intolerance has a genetic basis and why it has evolved differently in humans spread throughout the world.
- Lactose is the main sugar found in milk.
- Lactose intolerance is the inability to properly digest lactose.
- Instead of lactose being broken down and absorbed in the small intestine (小肠), lactose is broken down by bacteria in the large intestine, producing gas and discomfort.

Lactase (乳糖酶), the Enzyme for Lactose Breakdown

- Lack of lactase in small intestine.
- Lactose intolerance can be addressed by
 - avoiding lactose-containing foods or
 - lactose-free foods (yogurt) or
 - consuming lactase pills along with food
- Lactose also in breast milk, major carbohydrate source for infants.
- Lactase available in infants' small intestine.

A Genetic Basis?



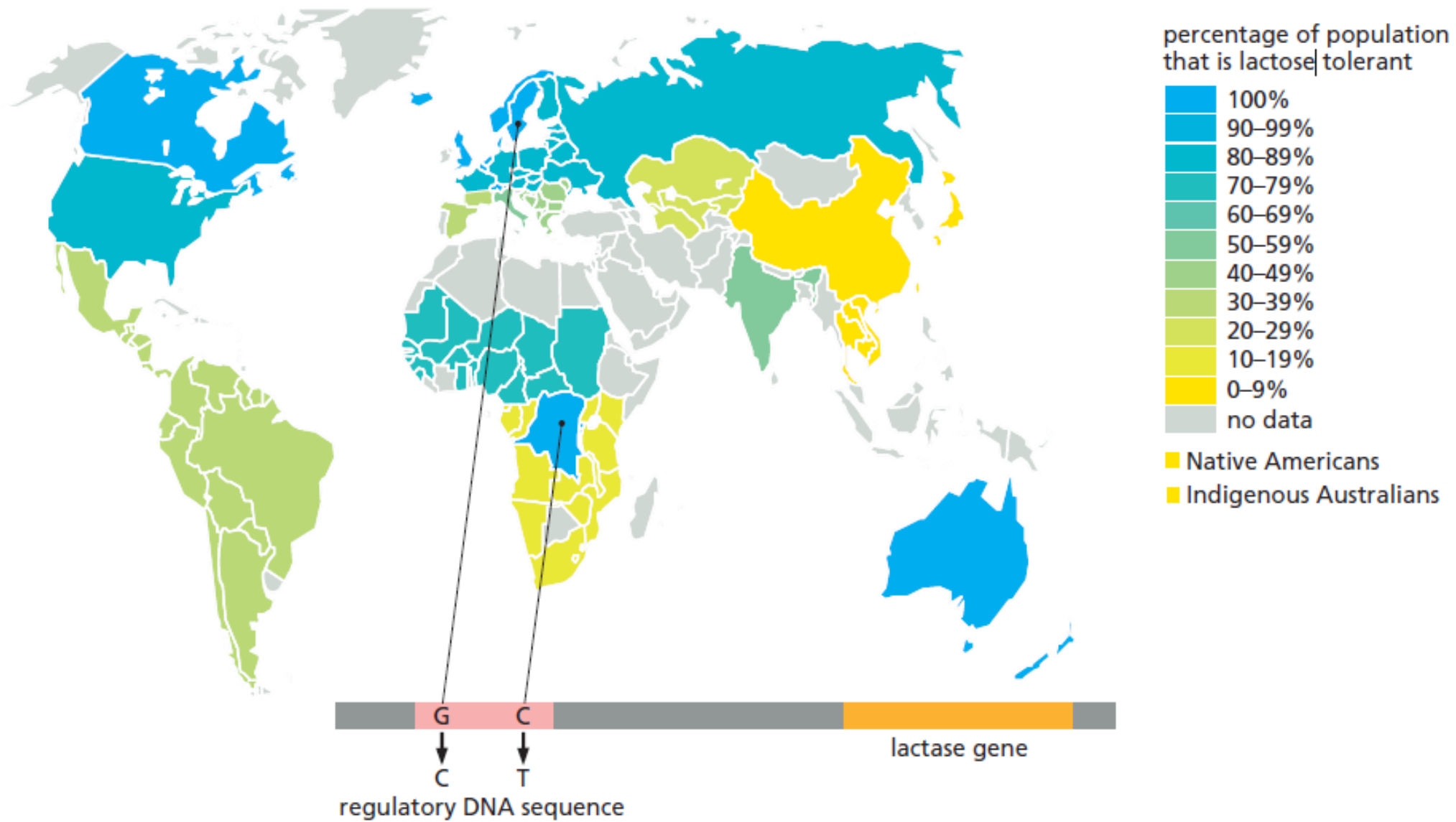


Figure 9–6 The widespread ability of adult humans to digest milk followed the domestication of cattle.